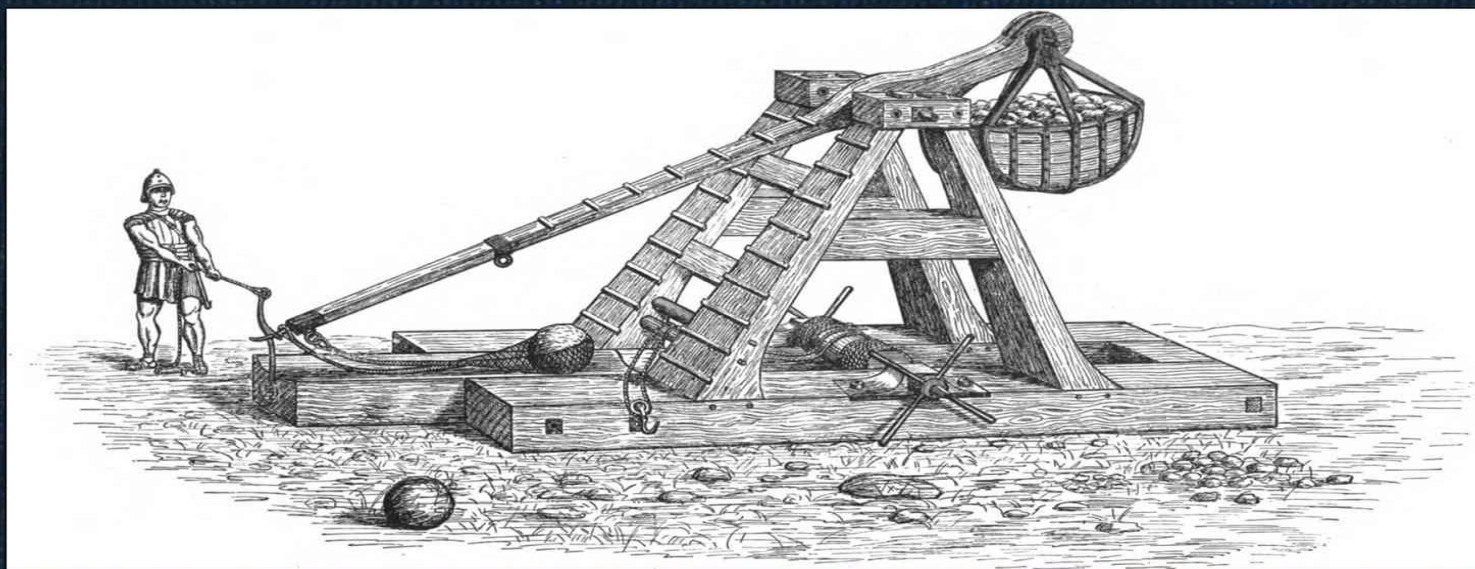


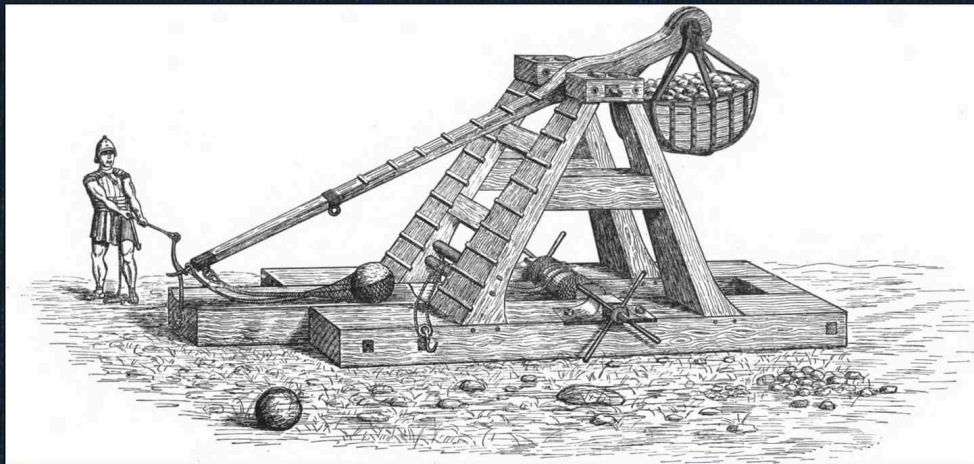
PAYNE-GALLWEY

SUMMARY OF THE HISTORY,
CONSTRUCTION AND EFFECTS
IN WARFARE OF THE
PROJECTILE-THROWING ENGINES
OF THE ANCIENTS ...



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**The Project Gutenberg eBook of A summary of the history,
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*** START OF THE PROJECT GUTENBERG EBOOK A SUMMARY
OF THE HISTORY, CONSTRUCTION AND EFFECTS IN WARFARE
OF THE PROJECTILE-THROWING ENGINES OF THE ANCIENTS,
WITH A TREATISE ON THE STRUCTURE, POWER AND
MANAGEMENT OF TURKISH AND OTHER ORIENTAL BOWS OF
MEDIÆVAL AND LATER TIMES ***

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[Additional notes](#) will be found near the end of this ebook.

**THE
PROJECTILE-THROWING
ENGINES
OF THE ANCIENTS
AND
TURKISH AND OTHER ORIENTAL
BOWS
OF MEDIÆVAL AND LATER TIMES**

By the same Author.

THE CROSSBOW,
MEDIÆVAL AND MODERN, MILITARY AND SPORTING:
Its Construction, History, and Management.
WITH A TREATISE ON
THE BALISTA AND CATAPULT OF THE ANCIENTS.

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OF THE ANCIENTS

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STRUCTURE, POWER AND MANAGEMENT
OF
TURKISH AND OTHER ORIENTAL BOWS
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BY
SIR RALPH PAYNE-GALLWEY, B^T.
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1907

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**THE PROJECTILE-THROWING
ENGINES OF
THE ANCIENTS**

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PREFACE

SINCE my recent book on mediæval archery and ancient weapons was issued,¹ I have obtained a considerable amount of information concerning the projectile engines of the Greeks and Romans. I now print a concise account of the history, construction and effects in warfare of these engines.

In this summary the additional notes I have acquired are included.

I also append a treatise fully describing that remarkable weapon the Turkish composite bow, which I only cursorily dealt with in the work referred to.

R. P. G.

THIRKLEBY PARK,
THIRSK:
Dec. 1906.

¹ *The Crossbow, Mediæval and Modern, Military and Sporting: its Construction, History, and Management. With a Treatise on the Balista and Catapult of the Ancients.* 220 illustrations. Messrs. Longmans & Co., 39 Paternoster Row, London.

PART I

INTRODUCTORY NOTES ON ANCIENT PROJECTILE ENGINES

OF ancient Greek authors who have left us accounts of these engines, Heron (284–221 B.C.) and Philo (about 200 B.C.) are the most trustworthy.

Both these mechanics give plans and dimensions with an accuracy that enables us to reconstruct the machines, if not with exactitude at any rate with sufficient correctness for practical application.

Though in the books of Athenæus, Biton, Apollodorus, Diodorus, Procopius, Polybius and Josephus we find incomplete descriptions, these authors, especially Josephus, frequently allude to the effects of the engines in warfare; and scanty as is the knowledge they impart, it is useful and explanatory when read in conjunction with the writings of Heron and Philo.

Among the Roman historians and military engineers, Vitruvius and Ammianus are the best authorities.

Vitruvius copied his descriptions from the Greek writers, which shows us that the Romans adopted the engines from the Greeks.

Of all the old authors who have described the engines, we have but copies of the original writings. It is therefore natural that we should come across many phrases and drawings which are evidently incorrect, as a result of repeated transcription, and which we know to be at fault though we cannot actually prove them to be so.

With few exceptions, all the authors named simply present us with their own ideas when they are in doubt respecting the mechanical details and performances of the engines they wish to describe.

All such spurious information is, of course, more detrimental than helpful to our elucidation of their construction and capabilities.

It frequently happens that in a mediæval picture of one of these machines some important mechanical detail is omitted, or, from the difficulty of portraying it correctly, is purposely concealed by figures of soldiers, an omission that may be supplied by reference to other representations of the same weapon.

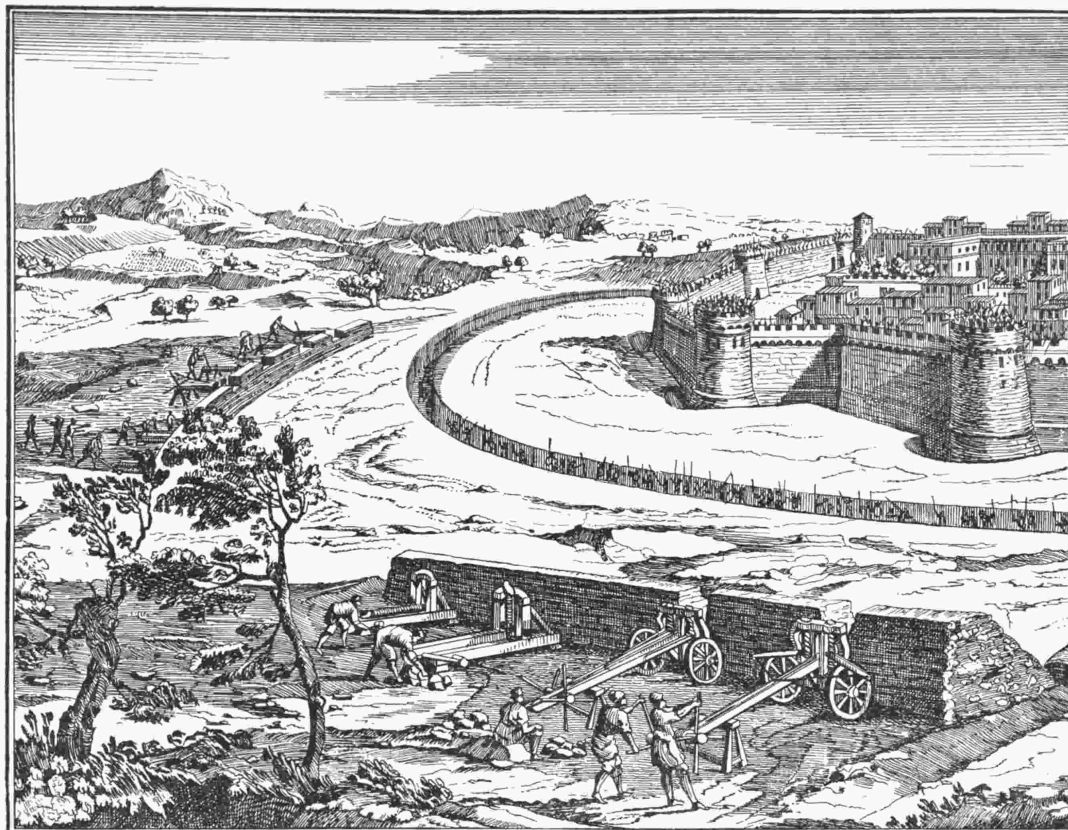


FIG. 1.—BESIEGING A FORTIFIED TOWN WITH A BATTERY OF CATAPULTS AND BALISTAS.

Criticism.—In this picture the balistas are fairly correct, but the catapults are too small.

From Polybius. Edition 1727.

It is, indeed, impossible to find a complete working plan of any one of these old weapons, a perfect design being only obtainable by consulting many

ancient authorities, and, it may be said, piecing together the details of construction they individually give.

* * * * *

We have no direct evidence as to when the engines for throwing projectiles were invented.

It does not appear that King Shalmaneser II. of Assyria (859–825 B.C.) had any, for none are depicted on the bronze doors of the palace of Balâwat, now in the British Museum, on which his campaigns are represented, though his other weapons of attack and defence are clearly shown.

The earliest allusion is the one in the Bible, where we read of Uzziah, who reigned from B.C. 808–9 to B.C. 756–7. ‘Uzziah made in Jerusalem engines invented by cunning men, to be on the towers and upon the bulwarks, to shoot arrows and great stones withal.’ (2 Chronicles xxvi. 15.)

Diodorus tells us that the engines were first seen about 400 B.C., and that when Dionysius of Syracuse organised his great expedition against the Carthaginians (397 B.C.) there was a genius among the experts collected from all over the world, and that this man designed the engines that cast stones and javelins.

From the reign of Dionysius and for many subsequent centuries, or till near the close of the fourteenth, projectile-throwing engines are constantly mentioned by military historians.

But it was not till the reign of Philip of Macedon (360–336 B.C.) and that of his son Alexander the Great (336–323 B.C.) that their improvement was carefully attended to and their value in warfare fully recognised.

As before stated, the Romans adopted the engines from the Greeks.

Vitruvius and other historians tell us this, and even copy their descriptions of them from the Greek authors, though too often with palpable inaccuracy.

To ascertain the power and mechanism of these ancient engines a very close study of all the old authors who wrote about them is essential, with a view to extracting here and there useful facts amid what are generally verbose and confused references.

There is no doubt that the engines made and used by the Romans after their conquest of Greece (B.C. 146), in the course of two or three centuries

became inferior to the original machines previously constructed by the Greek artificers.

Their efficiency chiefly suffered because the art of manufacturing their important parts was gradually neglected and allowed to become lost.

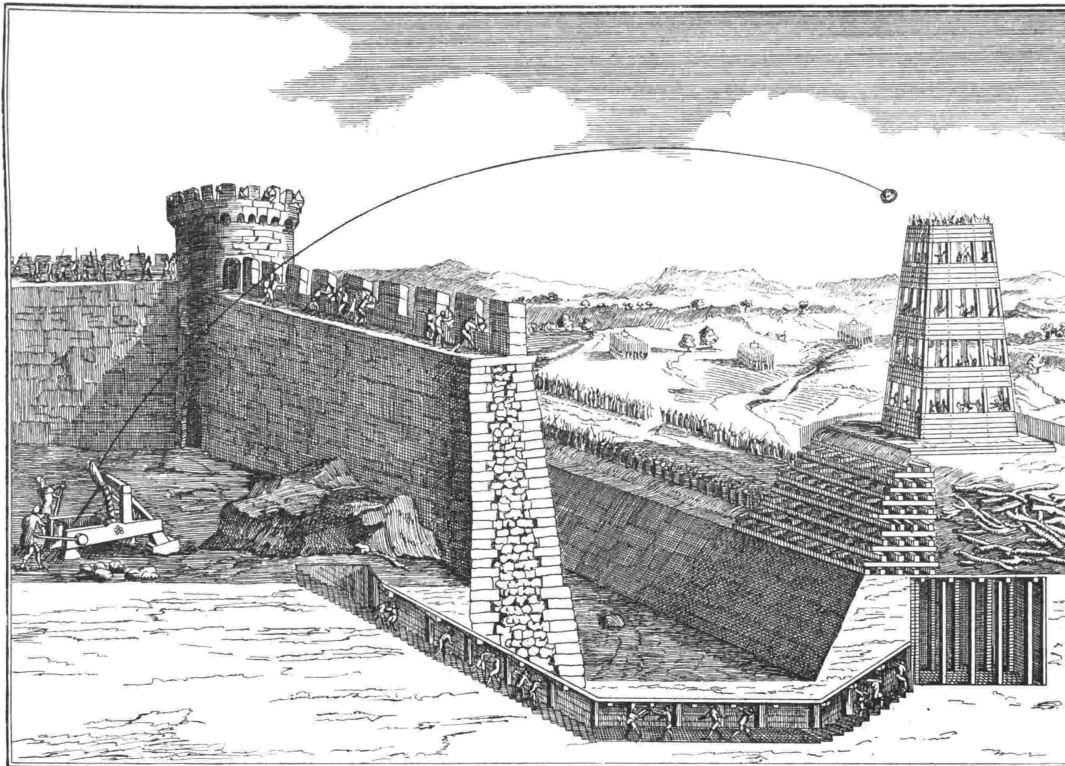


FIG. 2.—A SIEGE.

Criticism.—The picture is open to the spectator in order that he may see both defenders and besiegers at work.

The besieged have just cast a stone from a catapult. The stone is falling on the movable tower belonging to the attacking side.

From Polybius. Edition 1727.

For instance, how to make the skein of sinew that bestowed the very life and existence on every projectile-casting engine of the ancients.

The tendons of which the sinew was composed, the animals from which it was taken, and the manner in which it was prepared, we can never learn now.

Every kind of sinew, or hair or rope, with which I have experimented, either breaks or loses its elasticity in a comparatively short time, if great pressure is applied. It has then to be renewed at no small outlay of expense and trouble. Rope skeins, with which we are obliged to fit our models, cannot possibly equal in strength and above all in elasticity, skeins of animal sinew or even of hair.

The formation of the arm or arms of an engine, whether it is a catapult with its single upright arm or a balista with its pair of lateral ones, is another difficulty which cannot now be overcome, for we have no idea how these arms were made to sustain the great strain they had to endure.

We know that the arm of a large engine was composed of several spars of wood and lengths of thick sinew fitted longitudinally, and then bound round with broad strips of raw hide which would afterwards set nearly as hard and tight as a sheath of metal.

We know this, but we do not know the secret of making a light and flexible arm of sufficient strength to bear such a strain as was formerly applied to it in a catapult or a balista.

Certainly, by shaping an arm of great thickness we can produce one that will not fracture, but substance implies weight, and undue weight prevents the arm from acting with the speed requisite to cast its projectile with good effect.

A heavy and ponderous arm of solid wood cannot, of course, rival in lightness and effectiveness a composite one of wood, sinew and hide.

The former is necessarily inert and slow in its action of slinging a stone, while the latter would, in comparison, be as quick and lively as a steel spring.

When the art of producing the perfected machines of the Greeks was lost, they were replaced by less effective contrivances.

If the knowledge of constructing the great catapult of the ancients in its original perfection had been retained, such a clumsy engine as the mediæval trebuchet would never have gained popularity. The trebuchet derived its power from the gravity of an immense weight at one end of its pivoted arm tipping up the other end, to which a sling was attached for throwing a stone.

As regards range, there could be no comparison between the efficiency of a trebuchet, however large, as worked merely by a counterpoise, and that of an

engine deriving its power from the elasticity of an immense coil of tightly twisted sinew.

It is certain that if the latter kind of engine had survived in its perfect state the introduction of cannon would have been considerably delayed, for the effects in warfare of the early cannon were for a long period decidedly inferior to those of the best projectile engines of the ancients.

Notwithstanding many difficulties, I have succeeded in reconstructing, though of course on a considerably smaller scale, the chief projectile throwing engines of the ancients, and with a success that enables them to compare favourably, as regards their range, with the Greek and Roman weapons they represent.

Still, my engines are by no means perfect in their mechanism, and are, besides, always liable to give way under the strain of working.

One reason of this is that all modern engines of the kind require to be worked to their utmost capacity, *i.e.* to the verge of their breaking point, to obtain from them results that at all equal those of their prototypes.

A marked difference between the ancient engines and their modern imitations, however excellent the latter may be, is, that the former did their work easily, and well within their strength, and thus without any excessive strain which might cause their collapse after a short length of service.²

² Again, though my largest catapult will throw a stone to a great distance it cannot throw one of nearly the weight it should be able to do, considering the size of its frame, skein of cord and mechanism. In this respect it is decidedly inferior to the ancient engine.

The oft-disputed question as to the distance to which catapults and balistas shot their projectiles can be solved with approximate accuracy by comparing their performances—as given by ancient military writers—with the results obtainable from modern reproductions.

While treating of this matter we should carefully consider the position and surroundings of the engines when engaged in a siege, and especially the

work for which they were designed.

As an example, archers, with the advantage of being stationed on high towers and battlements, would be well able to shoot arrows from 270 to 280 yards. For this reason it was necessary for the safe manipulation of the attacking engines that they should be placed at about 300 yards from the outer walls of any fortress they were assailing.

As a catapult or a balista was required not only to cast its missile among the soldiers on the ramparts of a fortified place, but also to send it clear over the walls amid the houses and people within the defences, it is evident that the engines must have had a range of from 400 to 500 yards, or more, to be as serviceable and destructive as they undoubtedly were.

Josephus tells us that at the siege of Jerusalem, A.D. 70 ('Wars of the Jews,' Book V. Chapter VI.), stones weighing a talent ($57\frac{3}{4}$ lbs. avoirdupois) were thrown by the catapults to a distance of two or more 'stadest.'

This statement may be taken as trustworthy, for Josephus relates what he personally witnessed and his comments are those of a commander of high rank and intelligence.

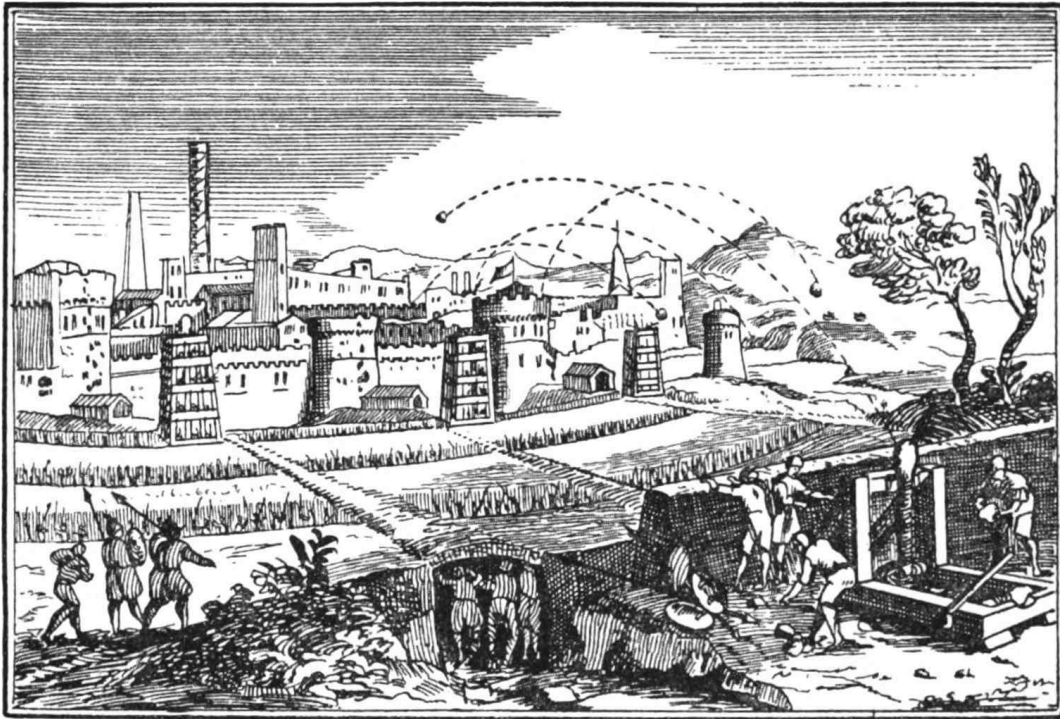


FIG. 3.—A FORTIFIED TOWN BEING BOMBARDED BY A CATAPULT.

Criticism.—The stones thrown by the besieged may be seen falling in the trenches of the besiegers. The catapult depicted is drawn on much too small a scale.

From Polybius. Edition 1727.

Two or more ‘stades,’ or let us say 2 to 2¼ ‘stades,’ represent 400 to 450 yards. Remarkable and conclusive testimony confirming the truth of what we read in Josephus is the fact that my largest catapult—though doubtless much smaller and less powerful than those referred to by the historian—throws a stone ball of 8 lbs. in weight to a range of from 450 to nearly 500 yards.

It is easy to realise that the ancients, with their great and perfect engines fitted with skeins of sinew, could cast a far heavier stone than one of 8 lbs., and to a longer distance than 500 yards.

Agesistratus,³ a Greek writer who flourished B.C. 200, and who wrote a treatise on making arms for war, estimated that some of the engines shot from 3½ to 4 ‘stades’ (700 to 800 yards).

³ The writings of Agesistratus are non-extant but are quoted by Athenæus.

Though such a very long flight as this appears almost incredible, I can adduce no sound reason for doubting its possibility. From recent experiments I am confident I could now build an engine of a size and power to accomplish such a feat if light missiles were used, and if its cost were not a consideration.

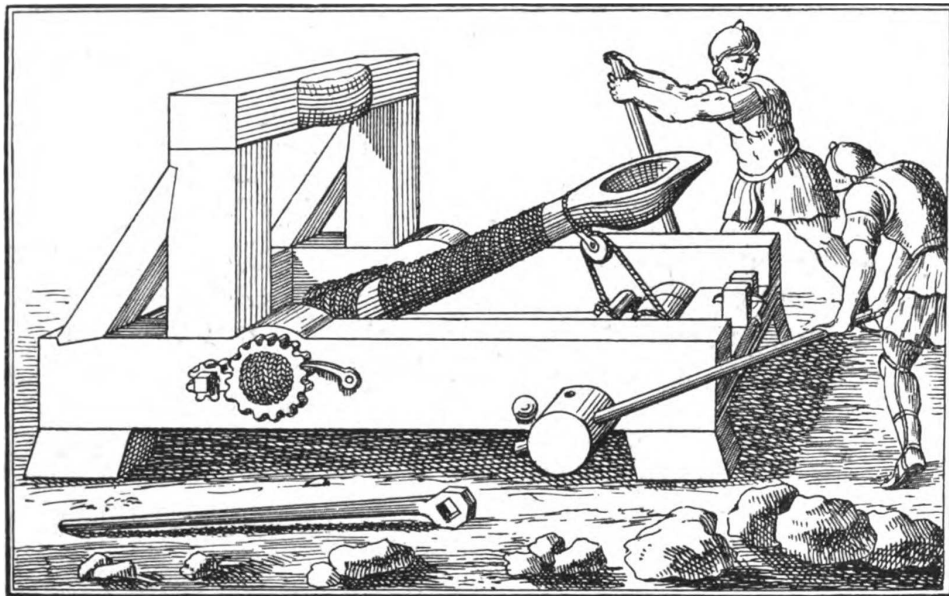


FIG. 4.—A SIEGE CATAPULT (WITHOUT A SLING).

From Polybius. Edition 1727.

PART II

THE CATAPULT (WITH A SLING)

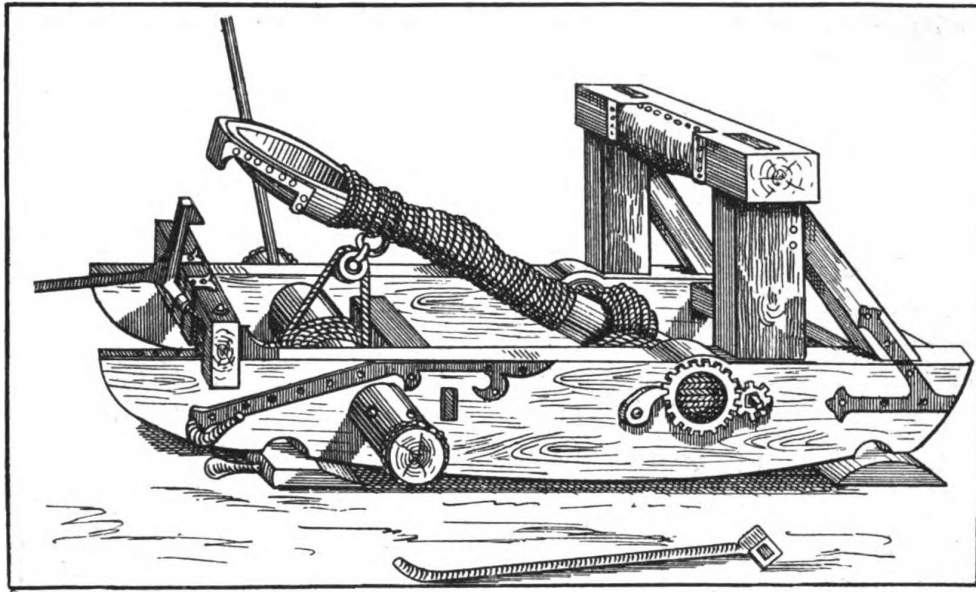


FIG. 5.—A SIEGE CATAPULT (WITHOUT A SLING).

Criticism.—This engine was moved into position on rollers and then props were placed under its sides to adjust the range of the projectile.

The end of the arm was secured by the notch of the large iron catch and was released by striking down the handle of the catch with a heavy mallet.

The arm is, however, too long for the height of the cross-bar against which it strikes and would probably break off at its centre.

The hollow for the stone is much too large, as a stone big enough to fit it could not be cast by a weapon of the dimensions shown in the picture.

From an Illustrated Manuscript, Fifteenth Century (No. 7239), Bibl. Nat. Paris.

THE mediæval catapult was usually fitted with an arm that had a hollow or cup at its upper end in which was placed the stone it projected, as shown above in [fig. 5](#).⁴ I find, however, that the original and more perfect form of

this engine, as employed by the Greeks and ancient Romans, had a sling, made of rope and leather, attached to its arm.⁵ ([Fig. 6](#), following page.)

⁴ See also *The Crossbow, etc.*, Chapters LV., LVI., illustrations 193 to 202.

⁵ In mediæval times catapults which had not slings cast great stones, but only to a short distance in comparison with the earlier weapons of the same kind that were equipped with slings. I can find no allusions or pictures to show that during this period any engine was used with a sling except the trebuchet, a post-Roman invention. All evidence goes to prove that the secret of making the skein and other important parts of a catapult was in a great measure lost within a couple of centuries after the Romans copied the weapon from their conquered enemies the Greeks, with the result that the trebuchet was introduced for throwing stones.

The catapult was gradually superseded as the art of its construction was neglected, and its efficiency in sieges was therefrom decreased.

The catapults of the fifth and sixth centuries were very inferior to those described by Josephus as being used at the sieges of Jerusalem and Jotapata (A.D. 70, A.D. 67), [p. 37](#).

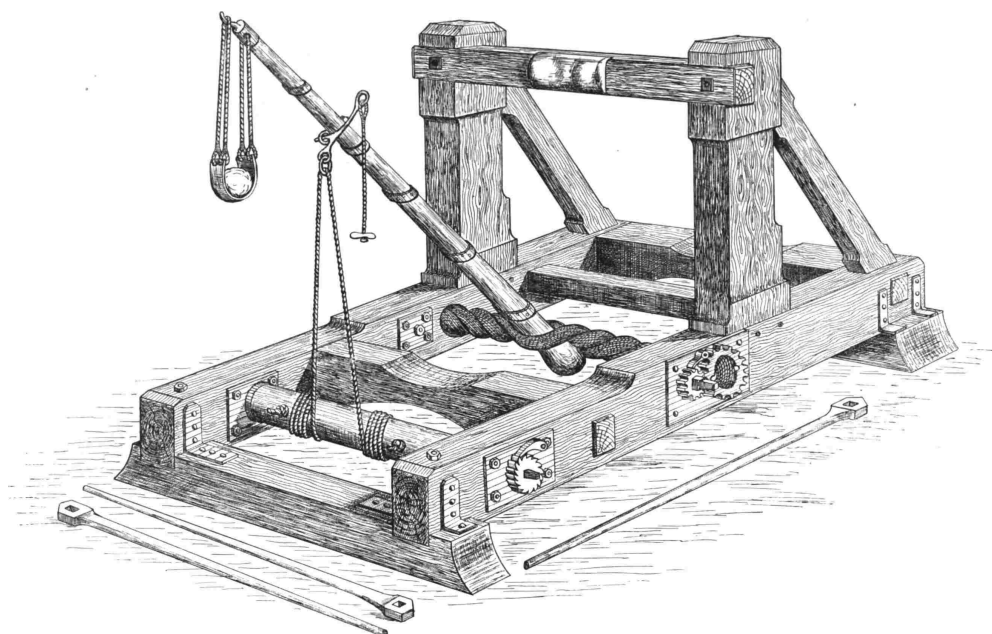


FIG. 6.—SKETCH PLAN OF A CATAPULT FOR SLINGING STONES ITS
ARM BEING PARTLY WOUND DOWN.

Approximate scale: $\frac{1}{4}$ in. = 1 ft.

The addition of a sling to the arm of a catapult increases its power by at least a third. For example, the catapult described in Chapters LV., LVI., of my book,⁶ will throw a round stone 8 lbs. in weight, from 350 to 360 yards, but the same engine with the advantage of a sling to its arm will cast the 8-lb. stone from 450 to 460 yards, and when its skein is twisted to its limit of tension to nearly 500 yards.

⁶ *The Crossbow, etc.*

If the upper end of the arm of a catapult is shaped into a cup to receive the stone, as shown in [fig. 5](#), p. 11, the arm is, of necessity, large and heavy at this part.

If, on the other hand, the arm is equipped with a sling, as shown in [fig. 6](#), opposite page, it can be tapered from its butt-end upwards, and is then much lighter and recoils with far more speed than an arm that has an enlarged extremity for holding its missile.

When the arm is fitted with a sling, it is practically lengthened by as much as the length of the sling attached to it, and this, too, without any appreciable increase in its weight.

The longer the arm of a catapult, the longer is its sweep through the air, and thus the farther will it cast its projectile, provided it is not of undue weight.

The difference in this respect is as between the range of a short sling and that of a long one, when both are used by a school-boy for slinging pebbles.

The increase of power conferred by the addition of a sling to the arm of a catapult is surprising.

A small model I constructed for throwing a stone ball, one pound in weight, will attain a distance of 200 yards when used with an arm that has a cup for holding the ball, though when a sling is fitted to the arm the range of the engine is at once increased to 300 yards.

The only historian who distinctly tells us that the catapult of the Greeks and Romans had a sling to its arm, is Ammianus Marcellinus. This author flourished about 380 A.D., and a closer study of his writings, and of those of his contemporaries, led me to carry out experiments with catapults and balistas which I had not contemplated when my work dealing with the projectile engines of the Ancients was published.

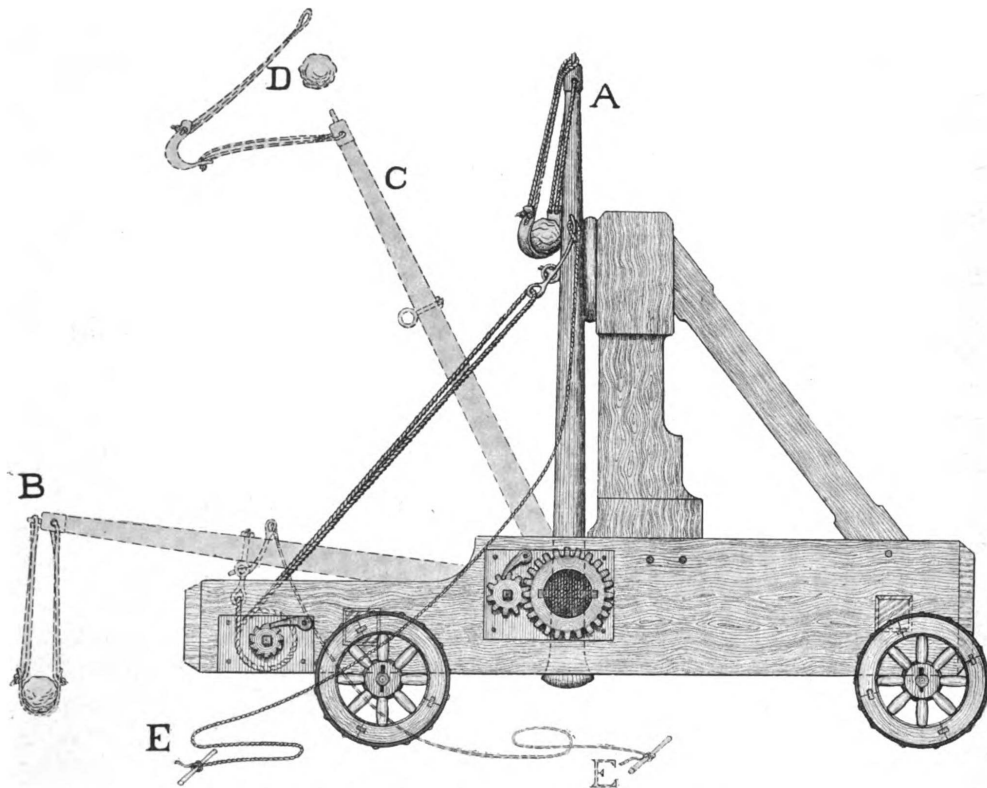


FIG. 7.—CATAPULT (WITH A SLING). Side view of frame and mechanism.

Scale: $\frac{1}{2}$ in. = 1 ft.

Ammianus writes of the catapult⁷:

‘In the middle of the ropes⁸ rises a wooden arm like a chariot pole ... to the top of the arm hangs a sling ... when battle is commenced a round stone is set in the sling ... four soldiers on each side of the engine wind the arm down till it is almost level with the ground ... when the arm is set free it springs up and hurls forth from its sling the stone, which is certain to crush whatever it strikes. This engine was formerly called the “scorpion,” because it has its sting erect,⁹ but later ages have given it the name of Onager, or wild ass, for when wild asses are chased they kick the stones behind them.’

⁷ *Roman History*, Book XXIII., Chapter IV.

⁸ *i.e.* in the middle of the twisted skein formed of ropes of sinew or hair.

⁹ The upright and tapering arm of a catapult, with the iron pin on its top for the loop of the sling, is here fancifully likened to the erected tail of an angry scorpion with its sting protruding.

FIG. 7.—CATAPULT (WITH A SLING), SEE OPPOSITE PAGE.

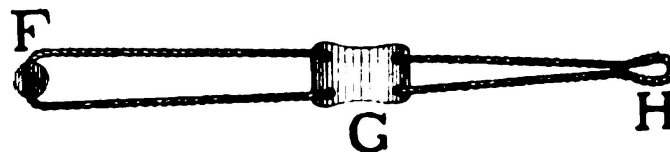
A. The arm at rest, ready to be wound down by the rope attached to it and also to the wooden roller of the windlass. The stone may be seen in the sling.

The upper end of the pulley rope is hitched by a metal slip-hook ([fig. 6](#), p. 12) to a ring-bolt secured to the arm just below the sling.

B. The position of the arm when fully wound down by means of the windlass and rope. See also EE, [fig. 8](#), page 16.

C. The position of the arm at the moment the stone D leaves the sling, which it does at an angle of about 45 degrees.

E. By pulling the cord E the arm B is at once released from the slip-hook and, taking an upward sweep of 90 degrees, returns to its original position at A.



THE SLING (OPEN).

[F. Its fixed end which passes through a hole near the top of the arm.

G. The leather pocket for the stone.

H. The loop which is hitched over the iron pin at the top of the arm when the stone is in position in the sling, as shown at A and B, [fig. 7](#).]

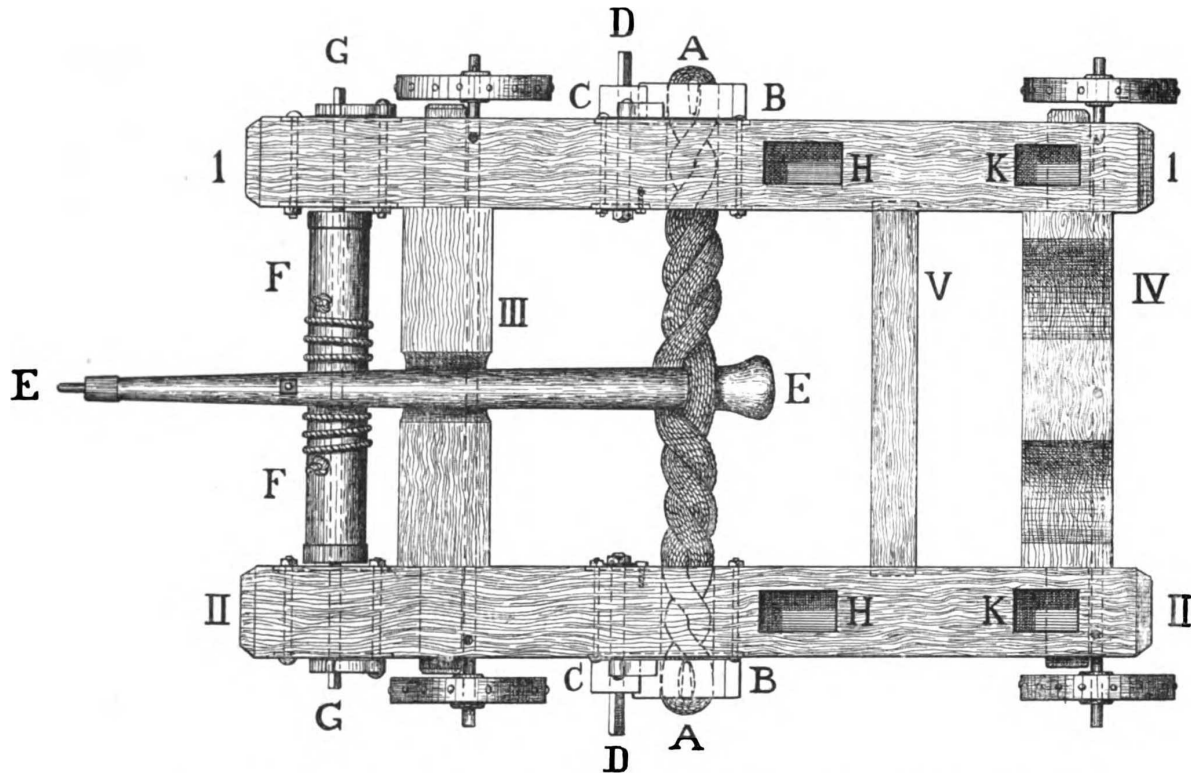


FIG. 8.—CATAPULT (WITH A SLING). Surface view of frame and mechanism. Scale: $\frac{1}{2}$ in. = 1 foot. The arm EE is here shown wound down to its full extent. (Compare with B, [fig. 7](#), page 14.)

- | | | |
|------|-----|-------------------------|
| I. | I. | } The side-pieces. |
| II. | II. | |
| III. | IV. | The large cross-pieces. |
| | V. | The small cross-piece. |

The ends of the cross-piece beams are stepped into the side-pieces.

AA. The skein of twisted cord.

BB. The large winding wheels. The skein is stretched between these wheels, its ends passing through the sides of the frame, and then through the wheels and over their cross-bars. ([Fig. 12](#), p. 19.)

By turning with a long spanner ([fig. 6](#), p. 12) the squared ends of the spindles DD, the pinion wheels CC rotate the large wheels BB and cause the latter to twist the skein AA, between the halves of which the arm EE is placed.

FF. The wooden roller which winds down the arm EE. ([Fig. 6](#), p. 12.)

The roller is revolved by four men (two on each side of the engine) who fit long spanners on the squared ends of the iron spindle GG.

This spindle passes through the centre of the roller and through the sides of the frame.

The small cogged wheels, with their checks, which are fitted to the ends of the spindle GG, prevent the roller from reversing as the arm is being wound down. ([Fig. 6](#), p. 12.)

HH. The hollows in the sides of the frame which receive the lower tenons of the two uprights. Between the tops of these uprights the cross-beam is fixed against which the arm of the catapult strikes when it is released. ([Fig. 6](#), p. 12.)

KK. The hollows for the lower tenons of the two sloping supports which prevent the uprights, and the cross-beam between them, from giving way when the arm recoils. ([Fig. 6](#), p. 12.)

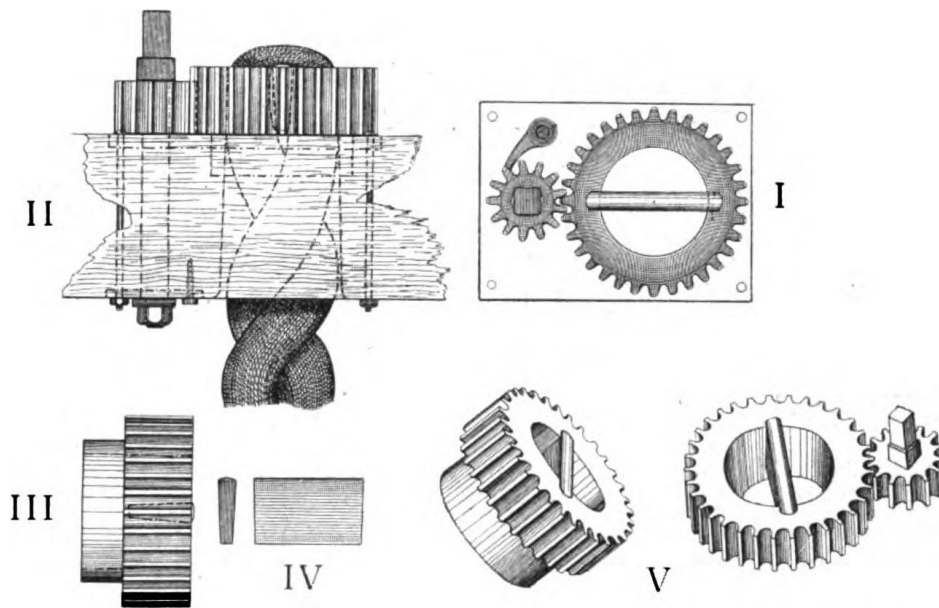


FIG. 9.—ONE OF THE PAIR OF WINCHES OF A CATAPULT. Scale:
1/16 in. = 1 in.

I. Surface view of one of the winches and of the thick iron plate in which the socket of the large winding wheel of the winch revolves.

II. View of a winch (from above) as fitted into one of the sides of the frame of the catapult. One end of the twisted skein may be seen turned round the cross-bar of the large wheel.

III. Side view of the large wheel of a winch.

IV. The cross-bar of one of the large wheels. These pieces fit like wedges into tapering slots cut down the barrels, or inside surfaces, of their respective wheels.

V. Perspective view of the wheels of a winch.

The winches are the vital parts of the catapult as they generate its projectile power.

They are employed to twist tightly the skein of cord between which the butt-end of the arm of the engine is placed.

The cord composing the skein is stretched to and fro across and through the sides of the catapult, and alternately through the insides of the large wheels and over their cross-bars; as shown in [fig. 8](#), p. 16.

[FIG. 10.](#) THE IRON SLIP-HOOK.

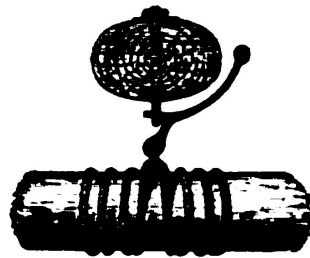


FIG. 10.

This simple contrivance not only pulled down the arm of a catapult but was also the means of setting it free. However great the strain on the slip-hook, it will, if properly shaped, easily effect the release of the arm.

The trajectory of the missile can be regulated by this form of release, as the longer the distance the arm is pulled down the higher the angle at which the projectile is thrown.

On the other hand, the shorter the distance the arm is drawn back, the lower the trajectory of its missile.

The slip-hook will release the arm of the engine at any moment, whether it is fully or only partially wound down by the windlass.

The slip-hook of the large catapult shown in [fig. 6](#), p. 12, has a handle, *i.e.* lever, 10 inches long, the point of the hook, which passes through the eye-bolt secured to the arm, being one inch in diameter.

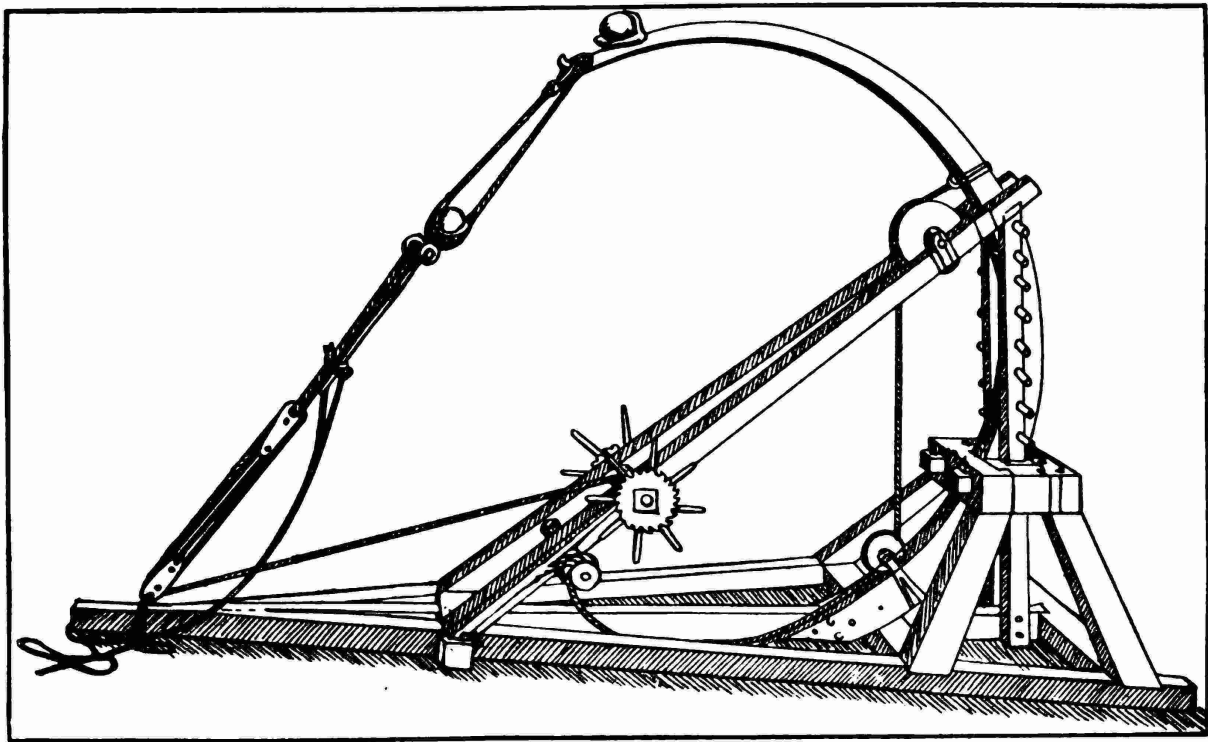


FIG. 11.—A SPRING ENGINE WITH A SLING ATTACHED TO ITS ARM, WHICH CAST TWO STONES AT THE SAME TIME.

From 'Il Codice Atlantico,' Leonardo da Vinci. 1445–1520.

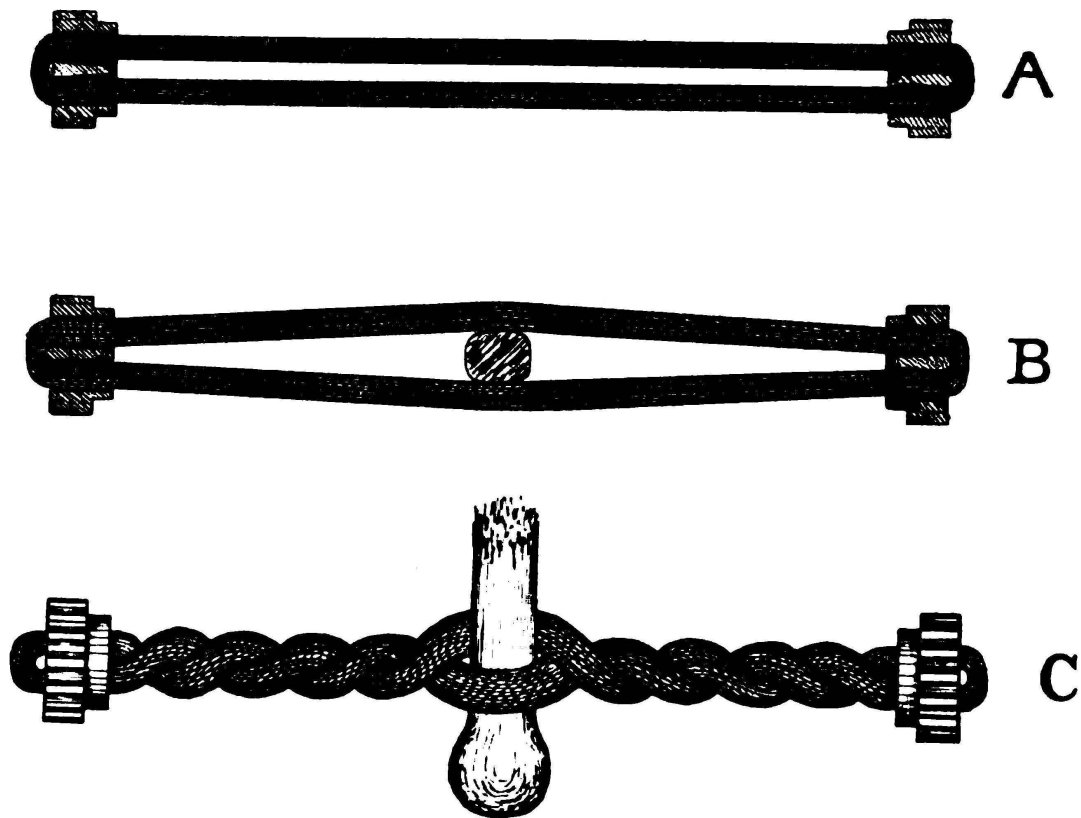


FIG. 12.—THE SKEIN OF CORD.

A. The skein as first wound over the cross-bars of the large wheels (shown in section) of the winches.

B. The skein with the butt-end of the arm (shown in section) placed between its halves.

C. The skein as it appears when tightly twisted up by the winches. Compare with AA, [fig. 8](#), p. 16.

Cord of Italian hemp, about $\frac{1}{4}$ in. thick, is excellent for small catapults. For large ones, horsehair rope, $\frac{1}{2}$ in. thick, is the best and most elastic. Whatever is used, the material of the skein must be thoroughly soaked in neats-foot oil for some days previously, or it is sure to fray and cut under the friction of being very tightly twisted. Oil will also preserve the skein from damp and decay for many years.

HOW TO WORK THE CATAPULT

There is little to write under this heading; as the plans, details of construction and illustrations will, I trust, elucidate its management.

The skein should never remain in a tightly twisted condition, but should be untwisted when the engine is not in use.

Previous to using the catapult its winches should be turned with the long spanner, [fig. 6](#), p. 12, first the winch on one side of the engine and then the one on the other side of it, and each to exactly the same amount.

Small numerals painted on the surfaces of the large wheels near their edges, will show how much they have been revolved; in this way their rotation can be easily arranged to correspond.

As the skein of cord is being twisted by the very powerful winches, the arm will gradually press with increasing force against the cross-beam between the uprights. The arm should be so tightly pressed against the fender, or cushion of straw, attached to the centre of this beam, that, whether large or small, it cannot be pulled back the least distance by hand.

If the skein of my largest catapult is fully tightened up by the winches, three strong men are unable to draw the arm back with a rope even an inch from the cross-beam, though the windlass has to pull it down from six to seven feet when the engine is made ready for action.

When the skein is as tight as it should be, attach the slip-hook to the ring-bolt in the arm and place the stone in the sling suspended from the top of the arm.

The arm can now be drawn down by means of long spanners fitted to the windlass. Directly the arm is as low as it should be, or as is desired, it should be instantly released by pulling the cord fastened to the lever of the slip-hook.

The least delay in doing this, and the resulting continuation of the immense strain on the arm, may cause it to fracture when it would not otherwise have done so.

The plans I have given are those of my largest engine, which, ponderous as it seems—it weighs two tons—is, however, less than half

the size of the catapult used by the ancients for throwing stones of from forty to fifty pounds in weight.

As the plans are accurately drawn to scale, the engine can easily be reproduced in a smaller size.

An interesting model can be constructed that has an arm 3 feet in length, and a skein of cord about 4 inches in diameter. It can be worked by one man and will throw a stone, the size of an orange, to a range of 300 yards.

The sling, when suspended with the stone in position, should be one third the length of the arm, as shown in [fig. 7](#), p. 14.

If the sling is shortened, the ball will be thrown at a high elevation. If the sling is lengthened, the ball will travel at a lower angle and with much more velocity.

PART III

THE BALISTA

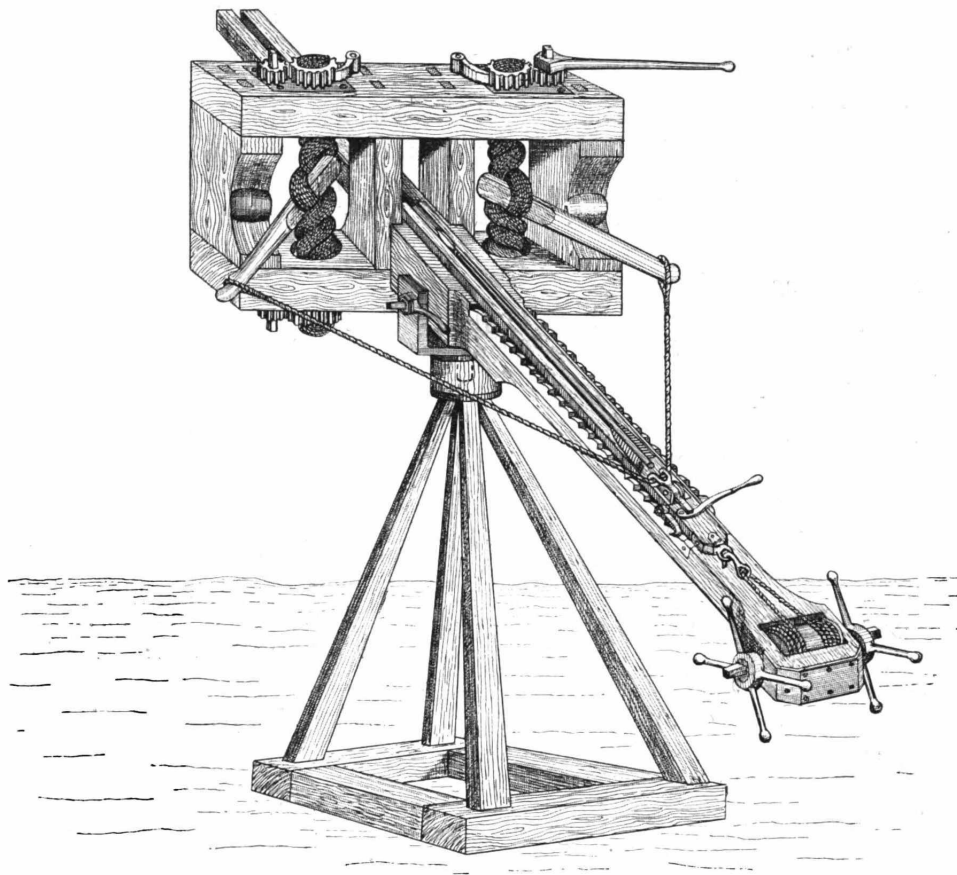


FIG. 13.—BALISTA FOR DISCHARGING HEAVY ARROWS OR
JAVELINS.

Approximate scale: $\frac{1}{2}$ in. = 1 foot.

THIS engine is here shown ready for discharge with its bow-string drawn to its full extent by the windlass.

The heavy iron-tipped arrow rests in the shallow wooden trough or groove which travels along the stock.

The trough has a strip of wood, in the form of a keel, fixed beneath it. This keel travels to or fro in a dove-tailed slot cut along the upper surface of the stock for the greater part of its length. (F, [fig. 14](#), p. 23.)

The arrow is laid in the trough before the bow-string is stretched. (A, B, [fig. 14](#), p. 23.)

The balista is made ready for use by turning the windlass. The windlass pulls back the sliding trough, and the arrow resting in it, along the stock of the engine, till the bow-string is at its proper tension for discharging the projectile. ([Fig. 13](#), p. 21.)

As the trough and the arrow are drawn back together, the arrow can be safely laid in position before the engine is prepared for action.

The catch for holding the bow-string, and the trigger for releasing it, are fixed to the solid after-end of the wooden trough. ([Fig. 14](#), p. 23.)

The two ratchets at the sides of the after-end of the trough travel over and engage, as they pass along, the metal cogs fixed on either side of the stock. ([Fig. 14](#), p. 23.)¹⁰

¹⁰ When the bow-string has been released and the arrow discharged, the ratchets are lifted clear of the cogs on the stock of the engine. This allows the trough to be slid forward to its first position as shown in A, B, [Fig. 14](#). It is then ready to be drawn back again for the next shot.

By this arrangement the trough can be securely retained, in transit, at any point between the one it started from and the one it attains when drawn back to its full extent by the windlass.

As the lock and trigger of the balista are fixed to the after-end of the sliding trough ([fig. 14](#), p. 23), it will be realised that the arrow could be discharged at any moment required in warfare, whether the bow-string was fully or only partially stretched.

In this respect the balista differed from the crossbow, which it somewhat resembled, as in a crossbow the bow-string cannot be set free by the trigger at an intermediate point, but only when it is drawn to the lock of the weapon.

It will be seen that the balista derives its power from two arms; each with its separate skein of cord and pair of winches.

These parts of the balista are the same in their action and mechanism as those of the catapult.

FIG. 14 (OPPOSITE PAGE).—THE MECHANISM OF THE STOCK OF AN ARROW-THROWING BALISTA.

A. Side view of the stock, with the arrow in the sliding trough before the bow-string is stretched.

B. Surface view of the stock, with the arrow in the sliding trough before the bow-string is stretched.

C. Section of the fore-end of the stock, and of the trough which slides in and along it.

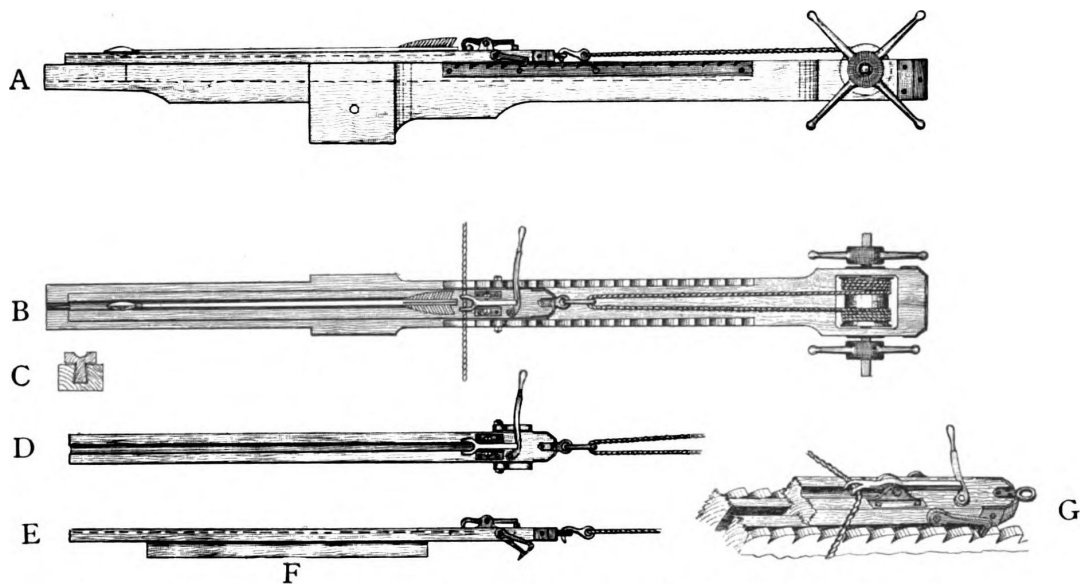


FIG. 14.—THE MECHANISM OF THE STOCK OF AN ARROW-THROWING BALISTA.

D. Surface view of the trough, with the trigger and catch for the bow-string.

E. Side view, showing the keel (F) which slides along the slot cut in the surface of the stock as the trough is drawn back by the windlass.

G. Enlarged view of the solid end of the trough. This sketch shows the catch for the bow-string, the trigger which sets it free, the ratchets which engage the cogs on the sides of the stock, and the slot cut in the stock for the dove-tailed keel of the trough to travel in.

* * * * *

Balistas were constructed of different sizes for the various purposes of siege and field warfare. The smallest of these engines was not much larger than a heavy crossbow, though it more than equalled the latter in power and range.

The small balistas were chiefly used for shooting through loopholes and from battlemented walls at an enemy assaulting with scaling ladders and movable towers.

The largest had arms of 3 ft. to 4 ft. in length, and skeins of twisted sinew of 6 in. to 8 in. in diameter.

Judging from models I have made and carefully experimented with; it is certain that the more powerful balistas of the ancients could cast arrows, or rather feathered javelins, of from 5 to 6 lbs. weight, to a range of from 450 to 500 yards.

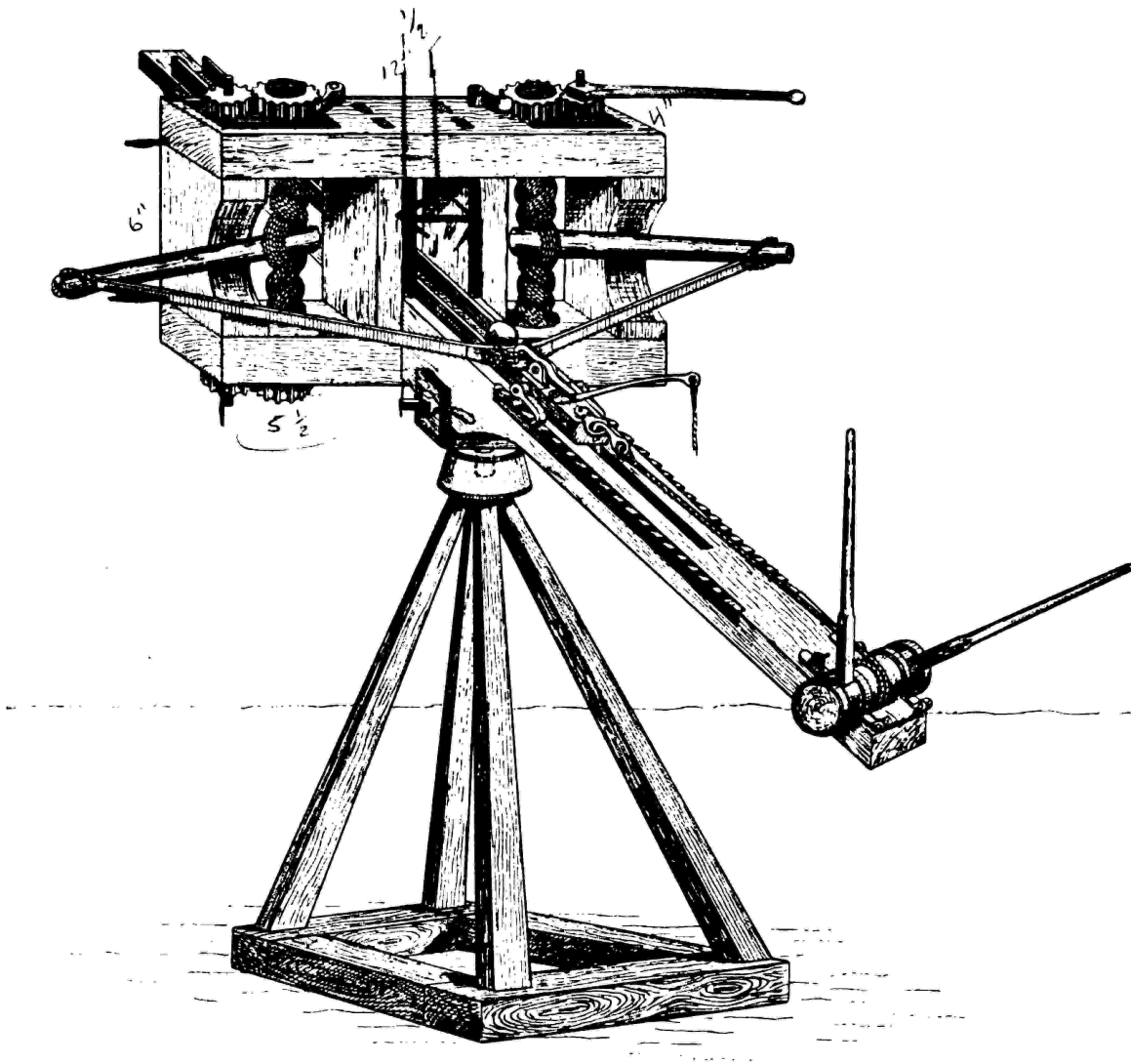


FIG. 15.—BALISTA FOR THROWING STONE BALLS. Approximate scale: $\frac{1}{2}$ in. = 1 foot.

This engine is here shown with its bow-string only slightly drawn along its stock by the windlass.

It will be seen that this engine is almost identical in construction with the one last described. ([Fig. 13](#), p. 21.)

The difference is that it propelled a stone ball instead of a large arrow.

The ball was driven along a square wooden trough, one-third of the diameter of the ball being enclosed by the sides of the trough so as to keep the missile in a true direction after the bow-string was released.

The bow-string was in the form of a broad band, with an enlargement at its centre against which the ball rested.

The description given of the mechanism and management of the engine for throwing arrows can be applied to the construction and manipulation of this form of balista, which was also made of large and small dimensions.

Small engines, with arms about 2 ft. in length and skeins of cord about 4 in. in diameter, such as those I have built for experiment, will send a stone ball, 1 lb. in weight, from 300 to 350 yards.

There is little doubt that the large stone-throwing balista of the Greeks and Romans was able to project a circular stone, of 6 to 8 lbs. weight, to a distance of from 450 to 500 yards.^{[11](#)}

¹¹ The balls used by the ancients in their catapults and balistas were often formed of heavy pebbles inclosed in baked clay, the reason being that balls made in this way shattered on falling and hence could not be shot back by the engines of the enemy. The balistas for throwing arrows, and those employed for casting stones, were fitted with axles and wheels when constructed for use in field warfare.

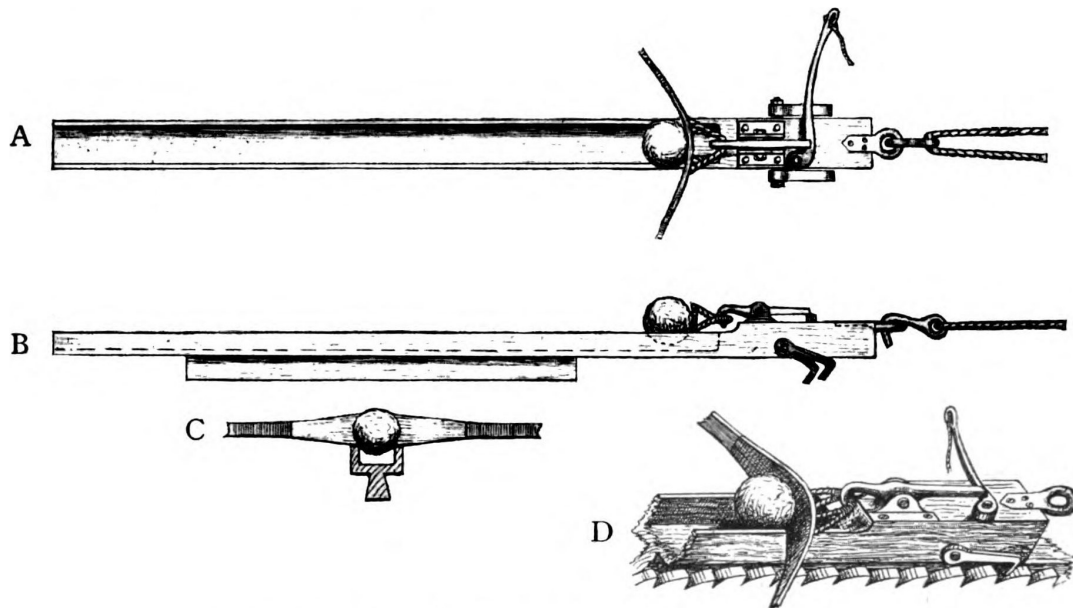


FIG. 16.—THE SLIDING TROUGH OF THE STONE-THROWING BALISTA.

A. Surface view, with the stone in position.

B. Side view, with the stone in position.

C. Front view of the stone as it rests in the trough against the enlarged centre of the bow-string.

D. Enlarged view of the solid end of the sliding trough. This sketch shows the ball in position against the bow-string; the catch holding the loop of the bow-string, and the pivoted trigger which, when pulled, releases the catch. One of the pair of ratchets which engage the cogs on the sides of the stock, as the trough is drawn back by the windlass to make ready the engine, is also shown. The trough has a keel to it, and slides to or fro along the stock in the same manner as in the arrow-throwing balista. ([Fig. 13](#), p. 21.)

Compare with [figs. 13, 14](#), pp. 21, 23, for further explanation of details.

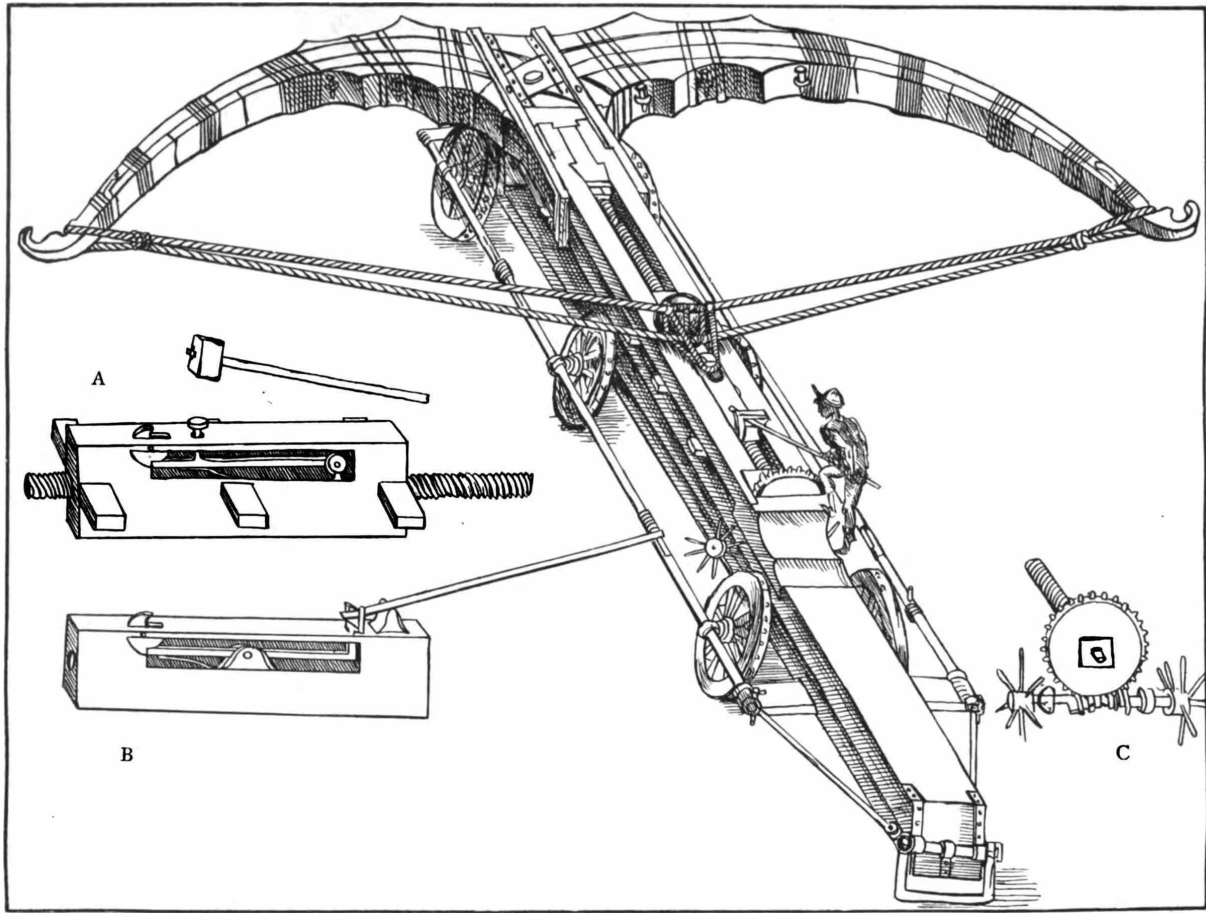


FIG. 17.—A SIEGE BALISTA IN THE FORM OF AN IMMENSE STONEBOW.

From 'Il Codice Atlantico,' Leonardo da Vinci, 1445–1520.

Criticism.—A stonebow of vast size. A and B represent two kinds of lock. In A, the catch of the lock over which the loop of the bow-string was hitched, was released by striking down the knob to be seen below the mallet. In B, the catch was set free by means of a lever. C shows the manner of pulling back the bow-string. By turning the spoked wheels, the screw-worm revolved the screwed bar on which the lock A, travelled. The lock, as may be seen, worked to or fro in a slot along the stock of the engine. In the illustration the bow is fully bent and the man indicated is about to discharge the engine. After this was done, the lock was wound back along the screw-bar and the bow-string was hitched over the catch of the lock preparatory to bending the bow again. Besides being a famous painter, Leonardo was distinguished as an inventor and exact writer on mechanics and hydraulics.

‘No artist before his time ever had such comprehensive talents, such profound skill or so discerning a judgment to explore the depths of every art or science to which he applied himself.’—JOHN GOULD, *Dictionary of Painters*, 1839.

From the above eulogy we may conclude that the drawings of ancient siege engines by Leonardo da Vinci are fairly correct.

PART IV

THE TREBUCHET

THIS engine was of much more recent invention than the catapult or the balista of the Greeks and Romans. It is said to have been introduced into siege operations by the French in the twelfth century. On the other hand, the catapult and the balista were in use several centuries before the Christian Era. Egidio Colonna gives a fairly accurate description of the trebuchet, and writes of it, about 1280, as though it were the most effective siege weapon of his time.

The projectile force of this weapon was obtained from the gravitation of a heavy weight, and not from twisted cordage as in the catapult and balista.

From about the middle of the twelfth century, the trebuchet in great measure superseded the catapult. This preference for the trebuchet was probably due to the fact that it was able to cast stones of about 300 lbs. in weight, or five or six times as heavy as those which the largest catapults could project.¹²

¹² The catapult had, besides, become an inferior engine to what it was some centuries before the trebuchet was introduced, the art of its construction having been neglected.

The stones thrown by the siege catapults of the time of Josephus would no doubt destroy towers and battlements, as the result of the constant and concentrated bombardment of many engines. One huge stone of from 200 to 300 lbs., as slung from a trebuchet, would, however, shake the strongest defensive masonry.

The trebuchet was essentially an engine for destroying the upper part of the walls of a fortress, so that it might be entered by means of scaling ladders or in other ways. The catapult, by reason of its longer range, was of more service in causing havoc to the people and dwellings inside the defences of a town.

From experiments with models of good size and from other sources, I find that the largest trebuchets—those with arms of about 50 ft. in length and counterpoises of about 20,000 lbs.—were capable of slinging a stone from 200 to 300 lbs. in weight to a distance of 300 yards, a range of 350 yards being, in my opinion, more than these engines were able to attain.¹³

¹³ Egidio Colonna tells us that the trebuchet was sometimes made without a counterpoise, and that in such a case the arm of the engine was worked by a number of men pulling together instead of by a heavy weight. I cannot believe this, as however many men pulled at the arm of a trebuchet they could not apply nearly the force that would be conveyed by the gravitation of a heavy weight.

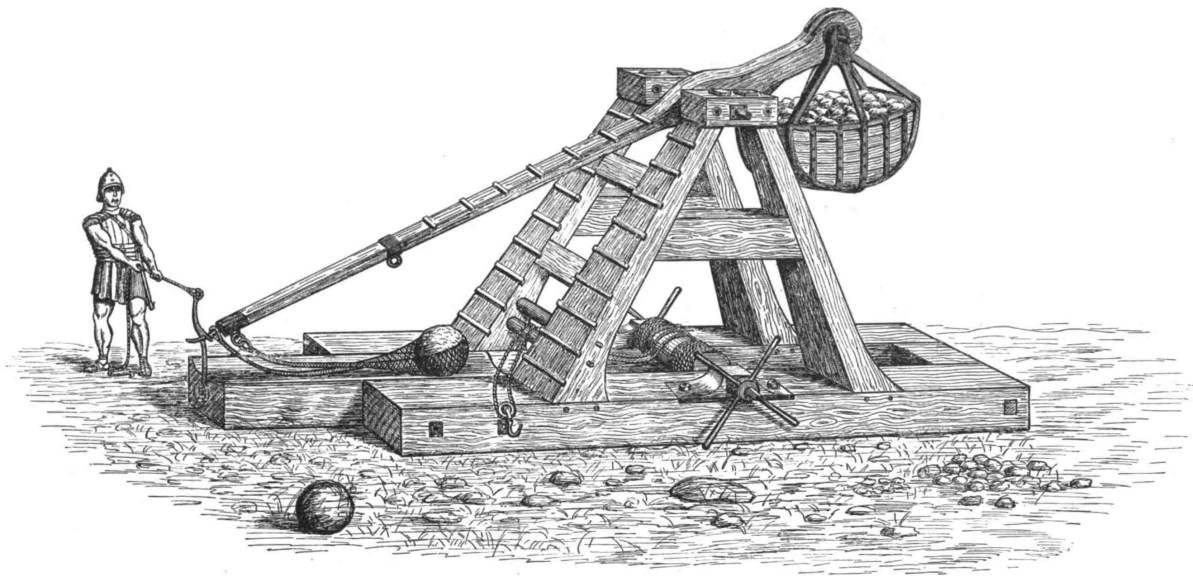


FIG. 18.—THE TREBUCHET.

The arm is fully wound down and the tackle of the windlass is detached from it. The stone is in the sling and the engine is about to be discharged by pulling the slip-hook off the end of the arm. The slip-hook is similar to the one shown in [fig. 10](#), p. 18.

N.B.—A Roman soldier is anachronistically shown in this picture. The trebuchet was invented after the time of the Romans.

The trebuchet always had a sling in which to place its missile.

The sling doubled the power of the engine and caused it to throw its projectile twice as far as it would have been able to do without it.

It was the length of the arm, when suitably weighted with its counterpoise, which combined with its sling gave power to the trebuchet. Its arm, when released, swung round with a long easy sweep and with nothing approaching the velocity of the much shorter arm of the catapult.

The weight of a projectile cast by a trebuchet was governed by the weight of its counterpoise. Provided the engine was of sufficient strength and could be manipulated, there was scarcely any limit to its power. Numerous references are to be found in mediæval authors to the practice of

throwing dead horses into a besieged town with a view to causing a pestilence therein, and there can be no doubt that trebuchets alone were employed for this purpose.

As a small horse weighs about 10 cwt., we can form some idea of the size of the rocks and balls of stone that trebuchets were capable of slinging.

When we consider that a trebuchet was able to throw a horse over the walls of a town, we can credit the statement of Stella,¹⁴ who writes 'that the Genoese armament sent against Cyprus in 1376 had among other great engines one which cast stones of 12 cwt.'

¹⁴ Stella flourished at the end of the fourteenth century and beginning of fifteenth. He wrote *The Annals of Genoa* from 1298–1409. Muratori includes the writings of Stella in his great work, *Rerum Italicarum Scriptores*, 25 vols., 1723–38.

Villard de Honnecourt¹⁵ describes a trebuchet that had a counterpoise of sand the frame of which was 12 ft. long, 8 ft. broad, and 12 ft. deep. That such machines were of vast size will readily be understood. For instance, twenty-four engines taken by Louis IX. at the evacuation of Damietta in 1249, afforded timber for stockading his entire camp.¹⁶ A trebuchet used at the capture of Acre by the Infidels in 1291, formed a load for a hundred carts.¹⁷ A great engine that cumbered the tower of St. Paul at Orleans and which was dismantled previous to the celebrated defence of the town against the English in 1428–9, furnished twenty-six cartloads of timber.¹⁸

¹⁵ Villard de Honnecourt, an engineer of the thirteenth century. His album translated and edited by R. Willis, M.A., 1859.

¹⁶ Jean, Sire de Joinville. He went with St. Louis to Damietta. His memoirs, written in 1309, published by F. Michel, 1858.

¹⁷ Abulfeda, 1273–1331. Arab soldier and historian, wrote *Annals of the Moslems*. Published by Hafnire, 1789–94. Abulfeda was himself in charge of one of the hundred carts.

¹⁸ From an old history of the siege (in manuscript) found in the town hall of Orleans and printed by Saturnin Holot, a bookseller of that city, 1576.

All kinds of articles besides horses, men, stones and bombs were at times thrown from trebuchets. Vassāf¹⁹ records ‘that when the garrison of Delhi refused to open the gates to Ala’uddin Khilji in 1296, he loaded his engines with bags of gold and shot them into the fortress, a measure which put an end to the opposition.’

¹⁹ Persian historian, wrote at end of thirteenth and beginning of fourteenth century. The preface to his history is dated 1288, and the history itself is carried down to 1312.

[Figs. 18, 20](#), pp. 28, 32, explain the construction and working of a trebuchet.

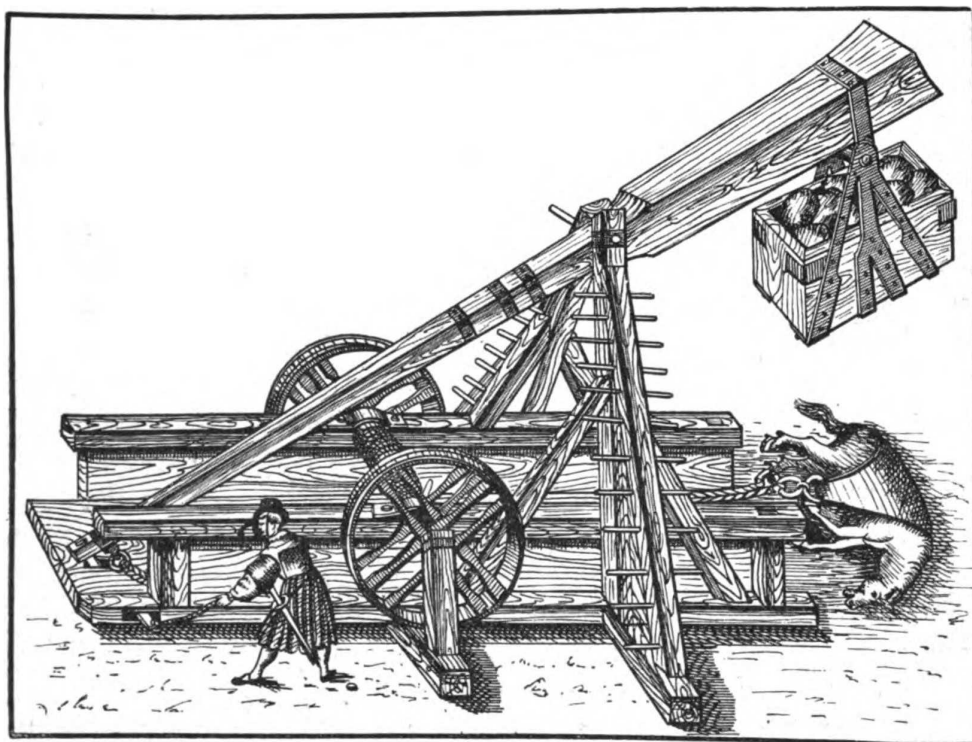


FIG. 19.—CASTING A DEAD HORSE INTO A BESIEGED TOWN BY
MEANS OF A TREBUCHET.

From 'Il Codice Atlantico,' Leonardo da Vinci, 1445–1520.

PART V

HISTORICAL NOTES ON ANCIENT AND MEDIÆVAL SIEGE ENGINES AND THEIR EFFECTS IN WARFARE

It is evident that a history of ancient siege engines cannot be created *de novo*. All that can be done is to quote with running criticism what has already been written about them.

The first mention of balistas and catapults is to be found in the Old Testament, two allusions to these weapons being made therein.

The references are:

2 Chronicles xxvi. 15, ‘And he²⁰ made in Jerusalem engines, invented by cunning men, to be on the towers and upon the bulwarks, to shoot arrows and great stones withal.’

²⁰ Uzziah.

Ezekiel xxvi. 9, ‘And he shall set engines of war against thy walls.’

Though the latter extract is not so positive in its wording as the one first given, it undoubtedly refers to engines that cast either stones or arrows against the walls, especially as the prophet previously alludes to other means of assault.

One of the most authentic descriptions of the use of great missive engines is to be found in the account by Plutarch of the siege of Syracuse by the Romans, 214–212 B.C.

Cæsar in his Commentaries on the Gallic and Civil wars, B.C. 58–50, frequently mentions the engines which accompanied him in his expeditions.

The balistas on wheels were harnessed to mules and called carro-balistas.

The carro-balista discharged its heavy arrow over the head of the animal to which the shafts of the engine were attached. Among the ancients these carro-balistas acted as field artillery and one is plainly shown in use on Trajan's Column.

According to Vegetius, every cohort was equipped with one catapult and every century with one carro-balista; eleven soldiers being required to work the latter engine.

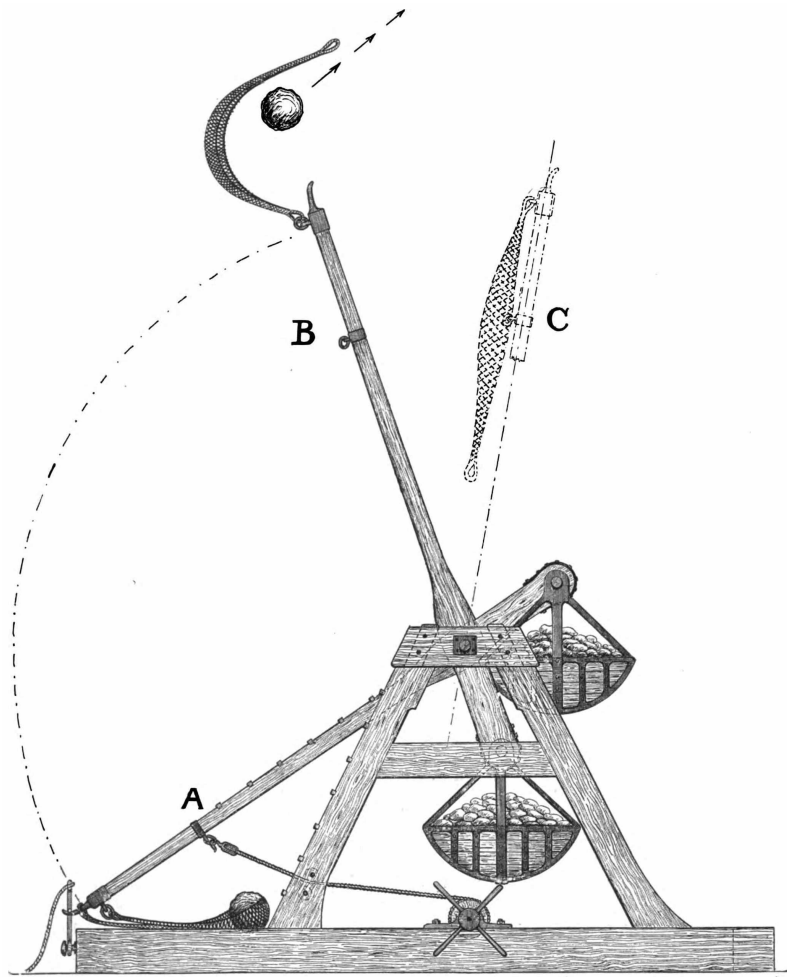


FIG. 20.—THE ACTION OF THE TREBUCHET.

A. The arm pulled down and secured by the slip-hook previous to unhooking the rope of the windlass. B. The arm released from the slip-hook and casting the stone out of its sling. C. The arm at the end of its upward sweep.

Sixty carro-balistas accompanied, therefore, besides ten catapults, a legion. The catapults were drawn along with the army on great carts yoked to oxen.

In the battles and sieges sculptured on Trajan's Column there are several figures of balistas and catapults. This splendid monument was erected in Rome, 105–113, to commemorate the victories of Trajan over the

Dacians, and constitutes a pictorial record in carved stone containing some 2,500 figures of men and horses.

It is astonishing what a large number of catapults and balistas were sometimes used in a siege. For instance, at the conquest of Carthage, B.C. 146, 120 great catapults and 200 small ones were taken from the defenders, besides 33 great balistas and 52 small ones (Livy).²¹

²¹ Just previous to the famous defence of Carthage, the Carthaginians surrendered to the Romans 'two hundred thousand suits of armour and a countless number of arrows and javelins, besides catapults for shooting swift bolts and for throwing stones to the number of two thousand.' From Appian of Alexandria, a Greek writer who flourished 98–161.

Abulfaragio (Arab historian, 1226–1286) records that at the siege of Acre in 1191, 300 catapults and balistas were employed by Richard I. and Philip II.

Abbo, a monk of Saint Germain des Prés, in his poetic but very detailed account of the siege of Paris by the Northmen in 885, 886, writes 'that the besieged had a hundred catapults on the walls of the town.'²²

²² These were probably balistas, as Ammianus Marcellinus writes of the catapult, 'An engine of this kind placed on a stone wall shatters whatever is beneath it, not by its weight but by the violence of its shock when discharged.'

Among our earlier English kings Edward I. was the best versed in projectile weapons large and small, including crossbows and longbows.

In the Calendar of Documents relating to Scotland, an account is given of his 'War-wolf,' a siege engine in the construction of which he was much

interested and which was no doubt a trebuchet.

This machine was of immense strength and size, and took fifty carpenters and five foremen a long time to complete. Edward designed it for the siege of Stirling, whither its parts were sent by land and by sea.

Sir Walter de Bedewyne, writing to a friend on July 20, 1304 (see Calendar of State Documents relating to Scotland), says: ‘As for news, Stirling Castle was absolutely surrendered to the King without conditions this Monday, St. Margaret’s Day, but the King wills it that none of his people enter the castle till it is struck with his “War-wolf,” and that those within the castle defend themselves from the said “War-wolf” as best they can.’

From this it is evident that Edward, having constructed his ‘War-wolf’ to cast heavy stones into the castle of Stirling to induce its garrison to surrender, was much disappointed by their capitulation before he had an opportunity of testing the power of his new weapon.

One of the last occasions on which the trebuchet was used with success is described by Guillet in his ‘Life of Mahomet II.’²³ This author writes: ‘At the siege of Rhodes in 1480, the Turks set up a battery of sixteen great cannon, but the Christians successfully opposed the cannon with a counter-battery of new invention.’²⁴

²³ Guillet de Saint George, born about 1625, died 1705. His *Life of Mahomet II.* was published in 1681. He was the author of several other works, including one on riding, warfare and navigation, termed the *Gentleman’s Dictionary*. The best edition of this book is in English and has many very curious illustrations. It is dated 1705.

²⁴ Called a new invention because the old siege engine of which this one (probably a trebuchet) was a reproduction had previously been laid aside for many years.

‘An engineer, aided by the most skilful carpenters in the besieged town, made an engine that cast pieces of stone of a terrible size. The execution wrought by this engine prevented the enemy from pushing forward the work of their approaches, destroyed their breastworks, discovered their mines, and filled with carnage the troops that came within range of it.’

At the siege of Mexico by Cortes in 1521, when the ammunition for the Spanish cannon ran short, a soldier with a knowledge of engineering undertook to make a trebuchet that would cause the town to surrender. A huge engine was constructed, but on its first trial the rock with which it was charged instead of flying into the town ascended straight upwards, and falling back to its starting-point destroyed the mechanism of the machine itself.²⁵

²⁵ *Conquest of Mexico*. W. Prescott, 1843.

Though all the projectile engines worked by cords and weights disappeared from continental warfare when cannon came to the front in a more or less improved form, they—if Vincent le Blanc is to be credited—survived in barbaric nations long after they were discarded in Europe.

This author (in his travels in Abyssinia) writes ‘that in 1576 the Negus attacked Tamar, a strong town defended by high walls, and that the besieged had engines composed of great pieces of wood which were wound up by cords and screwed wheels, and which unwound with a force that would shatter a vessel, this being the cause why the Negus did not assault the town after he had dug a trench round it.’²⁶

²⁶ Vincent le Blanc, *Voyages aux quatre parties du monde, redigé par Bergeron*, Paris, 1649. Though the accounts given by this author of his travels are imaginative, I consider his allusion to the siege engine to be trustworthy, as he was not likely to invent so correct a description of one.

Plutarch, in his Life of Marcellus the Roman General, gives a graphic account of Archimedes and the engines this famous mathematician employed in the defence of Syracuse.

It appears that Archimedes showed his relative Hiero II., King of Syracuse, some wonderful examples of the way in which immense weights could be moved by a combination of levers.

Hiero, being greatly impressed by these experiments, entreated Archimedes temporarily to employ his genius in designing articles of practical use, with the result that the scientist constructed for the king all manner of engines suitable for siege warfare.

Though Hiero did not require the machines, his reign being a peaceful one, they proved of great value shortly after his death when Syracuse was besieged by the Romans under Marcellus, 214–212 B.C.

On this occasion Archimedes directed the working of the engines he had made some years previously for Hiero.

Plutarch writes: ‘And in truth all the rest of the Syracusans were no more than the body in the batteries of Archimedes, whilst he was the informing soul. All other weapons lay idle and unemployed, his were the only offensive and defensive arms of the city.’

When the Romans appeared before Syracuse, its citizens were filled with terror, for they imagined they could not possibly defend themselves against so numerous and fierce an enemy.

But, Plutarch tells us, ‘Archimedes soon began to play his engines upon the Romans and their ships, and shot against them stones of such an enormous size and with so incredible a noise and velocity that nothing could stand before them. The stones overturned and crushed whatever came in their way, and spread terrible disorder through the Roman ranks. As for the machine which Marcellus brought upon several galleys fastened together, called *sambuca*²⁷ from its resemblance to the musical instrument of that name; whilst it was yet at a considerable distance, Archimedes discharged at it a stone of ten talents’ weight and, after that, a second stone and then a third one, all of which striking it with an amazing noise and force completely shattered it.²⁸

²⁷ *Sambuca*. A stringed instrument with cords of different lengths like a harp. The machine which Marcellus brought to Syracuse was designed to lift his soldiers—in small parties at a time and in quick succession—over the battlements of the town, so that when their numbers inside it were sufficient they might open its gates to the besiegers. The soldiers were intended to be hoisted on a platform, worked up and down by ropes and winches. As the machine was likened to a harp, it is probable it had a huge curved wooden arm fixed in an erect position and of the same shape as the modern crane used for loading vessels. If the arm of the *sambuca* had been straight like a mast, it could not have swung its load of men over a wall. Its further resemblance to a harp would be suggested by the ropes which were employed for lifting the platform to the summit of the arm, these doubtless being fixed from the top to the foot of the engine.

²⁸ It is, I consider, impossible that Archimedes, however marvellous the power of his engines, was able to project a stone of ten Roman talents or nearly 600 lbs. in weight, to a considerable distance! Plutarch probably refers to the talent of Sicily, which weighed about 10 lbs. A stone of ten Sicilian talents, or say 100 lbs., could have been thrown by a catapult of great strength and size.

Though the trebuchet cast stones of from 200 lbs. to 300 lbs. and more, this weapon was not invented till long after the time of Archimedes.

‘Marcellus in distress drew off his galleys as fast as possible and sent orders to his land forces to retire likewise. He then called a council of war, in which it was resolved to come close up to the walls of the city the next morning before daybreak, for they argued that the engines of Archimedes, being very powerful and designed to act at a long distance, would discharge their projectiles high over their heads. But for this Archimedes had been prepared, for he had engines at his disposal which were constructed to shoot at all ranges. When, therefore, the Romans came close to the walls,

undiscovered as they thought, they were assailed with showers of darts, besides huge pieces of rock which fell as it were perpendicularly upon their heads, for the engines played upon them from every quarter.

‘This obliged the Romans to retire, and when they were some way from the town Archimedes used his larger machines upon them as they retreated, which made terrible havoc among them as well as greatly damaged their shipping. Marcellus, however, derided his engineers and said, “Why do we not leave off contending with this geometrical Briareus, who sitting at ease and acting as if in jest has shamefully baffled our assaults, and in striking us with such a multitude of bolts at once exceeds even the hundred-handed giant of fable?”

‘At length the Romans were so terrified that, if they saw but a rope or a beam projecting over the walls of Syracuse, they cried out that Archimedes was levelling some machine at them and turned their backs and fled.’

As Marcellus was unable to contend with the machines directed by Archimedes and as his ships and army had suffered severely from the effects of these stone- and javelin-casting weapons, he changed his tactics and instead of besieging the town he blockaded it and finally took it by surprise.

Though, at the time of the siege of Syracuse, Archimedes gained a reputation for divine rather than human knowledge in regard to the methods he employed in the defence of the city, he left no description of his wonderful engines, for he regarded them as mere mechanical appliances which were beneath his serious attention, his life being devoted to solving abstruse questions of mathematics and geometry.

Archimedes was slain at the capture of Syracuse, B.C. 212, to the great regret of Marcellus.

The following extracts from Josephus, as translated by Whiston, enable us to form an excellent idea of the effects of great catapults in warfare:

(1) *Wars of the Jews*, Book III., Chapter VII.—The siege of Jotapata, A.D. 67. ‘Vespasian then set the engines for throwing stones and darts round about the city; the number of the engines was in all a hundred and sixty.... At the same time such engines as were intended for that purpose threw their

spears buzzing forth, and stones of the weight of a talent were thrown by the engines that were prepared for doing so....

‘But still Josephus and those with him, although they fell down dead one upon another by the darts and stones which the engines threw upon them, did not desert the wall.... The engines could not be seen at a great distance and so what was thrown by them was hard to be avoided; for the force with which these engines threw stones and darts made them wound several at a time, and the violence of the stones that were cast by the engines was so great that they carried away the pinnacles of the wall and broke off the corners of the towers; for no body of men could be so strong as not to be overthrown to the last rank by the largeness of the stones.... The noise of the instruments themselves was very terrible, the sound of the darts and stones that were thrown by them was so also; of the same sort was that noise that dead bodies made when they were dashed against the wall.’

(2) *Wars of the Jews*, Book V., Chapter VI.—The siege of Jerusalem, A.D. 70. ‘The engines that all the legions had ready prepared for them were admirably contrived; but still more extraordinary ones belonged to the tenth legion: those that threw darts and those that threw stones were more forcible and larger than the rest, by which they not only repelled the excursions of the Jews but drove those away who were upon the walls also. Now the stones that were cast were of the weight of a talent²⁹ and were carried two or more stades.³⁰

²⁹ 57¾ lbs. (avoirdupois).

³⁰ Two stades would be 404 yards; the measure of a stade is 606¾ English feet.

‘The blow they gave was no way to be sustained, not only by those who stood first in the way but by those who were beyond them for a great space.

‘As for the Jews, they at first watched the coming of the stone, for it was of a white colour and could therefore not only be perceived by the great

noise it made, but could be seen also before it came by its brightness; accordingly the watchmen that sat upon the towers gave notice when an engine was let go ... so those that were in its way stood off and threw themselves down upon the ground. But the Romans contrived how to prevent this by blacking the stone; they could then aim with success when the stone was not discerned beforehand, as it had been previously.'

The accounts given by Josephus are direct and trustworthy evidence, for the reason that this chronicler relates what he personally witnessed during the sieges he describes, in one of which (Jotapata) he acted the part of a brave and resourceful commander.

Tacitus in describing a battle fought near Cremona between the armies of Vitellius and Vespasian, A.D. 69, writes: 'The Vitellians at this time changed the position of their battering-engines, which in the beginning were placed in different parts of the field and could only play at random against the woods and hedges that sheltered the enemy. They were now moved to the Postumian way, and thence having an open space before them could discharge their missiles with good effect.'³¹

³¹ Tacitus continues: 'The fifteenth legion had an engine of enormous size, which was played off with dreadful execution and discharged massy stones of a weight to crush whole ranks at once. Inevitable ruin must have followed if two soldiers had not signalled themselves by a brave exploit. Covering themselves with shields of the enemy which they found among the slain, they advanced undiscovered to the battering-engine and cut its ropes and springs. In this bold adventure they both perished and with them two names that deserved to be immortal.'

Froissart chronicles that at the siege of Thyn-l'Evêque, 1340, in the Low Countries, 'John, Duke of Normandy had a great abundance of engines carted from Cambrai and Douai. Among others he had six very large ones which he placed before the fortress, and which day and night cast great stones which battered in the tops and roofs of the towers and of the rooms

and halls, so much so that the men who defended the place took refuge in cellars and vaults.'

Camden records that the strength of the engines employed for throwing stones was incredibly great and that with the engines called mangonels³² they used to throw millstones. Camden adds that when King John laid siege to Bedford Castle, there were on the east side of the castle two catapults battering the old tower, as also two upon the south side besides another on the north side which beat two breaches in the walls.

³² Catapults were often called mangons or mangonels, but in course of time the name mangonel was applied to any siege engine that projected stones or arrows. In this case the trebuchet is intended, as no catapult could project a millstone.

The same authority asserts that when Henry III. was besieging Kenilworth Castle, the garrison had engines which cast stones of an extraordinary size, and that near the castle several balls of stone sixteen inches in diameter have been found which are supposed to have been thrown by engines with slings³³ in the time of the Barons' war.

³³ The engines here alluded to by Camden were trebuchets.

Holinshed writes that 'when Edward I. attacked Stirling Castle, he caused an engine of wood to be set up to batter the castle which shot stones of two or three hundredweight.' (See allusion to this, [p. 33](#).)

Père Daniel, in his *Histoire de la Milice Française*, writes: 'The great object of the French engineers was to make siege engines of sufficient strength to project stones large enough to crush in the roofs of houses and break down the walls.' This author continues: 'The French engineers were

so successful and cast stones of such enormous size that their missiles even penetrated the vaults and floors of the most solidly built houses.’³⁴

³⁴ These engines would also be trebuchets.

The effects of the balista on the defenders of a town were in no degree inferior to those of the catapult. The missile of the balista consisted of a huge metal-tipped wooden bolt which, although of far less weight than the great ball of stone cast by a catapult or the far larger one thrown by a trebuchet, was able to penetrate roofs and cause great destruction in ranks of soldiers. Cæsar records that when his lieutenant Caius Trebonius was building a movable tower at the siege of Marseilles, the only method of protecting the workmen from the darts of engines³⁵ was by hanging curtains woven from cable-ropes on the three sides of the tower exposed to the besiegers.³⁶

³⁵ Balistas.

³⁶ ‘For this was the only sort of defence which they had learned, by experience in other places, could not be pierced by darts or engines.’ Cæsar’s *Commentaries on the Civil War*, Book II., Chapter IX.

Procopius relates that during the siege of Rome in 537 by Vitiges King of Italy, he saw a Gothic chieftain in armour suspended to a tree which he had climbed, and to which he had been nailed by a balista bolt which had passed through his body and then penetrated into the tree behind him.

Again, at the siege of Paris by the Northmen in 885–886, Abbo writes that Ebolus³⁷ discharged from a balista a bolt which transfixed several of the enemy.

³⁷ Abbot of Saint-Germain des Prés and one of the chief defenders of the town.

With grim humour Ebolus bade their comrades carry the slain to the kitchen, his suggestion being that the men impaled on the shaft of the balista resembled fowls run through with a spit previous to being roasted.

Not only were ponderous balls of stone and heavy bolts projected into a town and against its walls and their defenders, but, with a view to causing a pestilence, it was also the custom to throw in dead horses, and even the bodies of soldiers who had been killed in sorties or assaults.

For example, Varillas³⁸ writes that ‘at his ineffectual siege of Carolstein in 1422, Coribut caused the bodies of his soldiers whom the besieged had killed to be thrown into the town in addition to 2,000 cartloads of manure. A great number of the defenders fell victims to the fever which resulted from the stench, and the remainder were only saved from death by the skill of a rich apothecary who circulated in Carolstein remedies against the poison which infected the town.’

³⁸ French historian, born 1624, died 1696.

Froissart tells us that at the siege of Auberoche, an emissary who came to treat for terms was seized and shot back into the town. This author writes:

‘To make it more serious, they took the varlet and hung the letters round his neck and instantly placed him in the sling of an engine and then shot him back again into Auberoche. The varlet arrived dead before the knights who were there and who were much astonished and discomfited when they saw him arrive.’

Another historian explains that to shoot a man from the sling of an engine he must first be tied up with ropes, so as to form a round bundle like

a sack of grain.

The engine with which such fiendish deeds were achieved was the trebuchet.

A catapult was not powerful enough to project the body of a man. This difficulty was overcome by cutting off the head of any unfortunate emissary for peace, if the terms he brought were scornfully rejected. His letter of supplication from the besieged was then nailed to his skull, and his head was sent flying through space to fall inside the town as a ghastly form of messenger conveying a refusal to parley.

As it was always an object to the besiegers of a town to start a conflagration if they could, Greek fire was used for the purpose. The flame of this fearfully destructive liquid, the composition of which is doubtful, could not be quenched by water. It was placed in round earthenware vessels that broke on falling, and which were shot from catapults; as the roofs of ancient and mediæval dwelling-houses were usually thatched, it of course dealt destruction when it encountered such combustible material.

The successful attack or defence of a fortified town often depended on which of the armies engaged had the more powerful balistas, catapults or trebuchets, as one engine of superior range could work destruction unimpeded if it happened that a rival of similar power was not available to check its depredations.

Froissart relates that 'at the siege of Mortagne in 1340, an engineer within the town constructed an engine to keep down the discharges of one powerful machine in the besieging lines. At the third shot he was so lucky as to break the arm of the attacking engine.' The account of this incident, as given by Froissart, is so quaint and graphic that I quote it here: 'The same day they of Valencens raysed on their syde a great engyn and dyd cast in stones so that it troubled sore them within the town. Thus ye firste day passed and the night in assayling and devysing how they might greve them in the fortress.

'Within Mortagne there was a connyng maister in making of engyns who saw well how the engyn of Valencens did greatly greve them: he raysed an engyn in ye castle, the which was not very great but he trymmed it to a point,³⁹ and he cast therwith but three tymes. The firste stone fell a

xii⁴⁰ fro the engyn without, the second fell on y^e engyn, and the thirde stone hit so true that it brake clene asonder the shaft of the engyn without; then the soldyers of Mortagne made a great shout, so that the Hainaulters could get nothing ther⁴¹; then the erle⁴² sayd how he wolde withdrawe.'

³⁹ i.e. with great exactness or 'to a hair.'

⁴⁰ A foot.

⁴¹ Could not throw any more stones.

⁴² Count of Hainault. He was besieging Tournay, but left that place and went to besiege Mortagne and ordered the people of Valenciennes to go with him.

(From the translation made at the request of Henry VIII. by John Bouchier, second Lord Berners, published 1523–1525.)

These siege engines when only of moderate size were not always successful, as in some cases the walls of a town were so massively built that the projectiles of the enemy made little impression upon them. Froissart tells us that it was then the habit of the defenders of the walls to pull off their caps, or produce cloths, and derisively dust the masonry when it was struck by stones.

Some of the historians, mechanics and artists from whom information on balistas, catapults and trebuchets may be derived, are as follows. I name them alphabetically irrespective of their periods:

ABBO: A monk of Saint-Germain des Prés, born about the middle of the ninth century, died in 923. He wrote a poem in Latin describing the siege of Paris by the Northmen in 885–886.

AMMIANUS MARCELLINUS: Military historian. Died shortly after 390. His work first printed at Rome 1474. The latest edition is that of

V. Gardthausen, 1874–1875.

APPIAN: Historian. Lived at Rome during the reigns of Trajan, Hadrian and Antoninus Pius, 98–161. The best edition of his History is that of Schweighaeuser, 1785.

APOLLODORUS OF DAMASCUS: Built Trajan's Column, 105–113. Architect and engineer. Addressed a series of letters to the Emperor Trajan on siege engines (*vide* Thévenot).

ATHENÆUS: Lived in the time of Archimedes, B.C. 287–212. The author of a treatise on warlike engines (*vide* Thévenot).

BITON: Flourished about 250 B.C. Wrote a treatise on siege engines for throwing stones (*vide* Thévenot).

BLONDEL, FRANÇOIS: French engineer and architect; born 1617; died 1686.

CÆSAR, JULIUS (the Dictator): Born B.C. 100; died B.C. 44. Author of the 'Commentaries' on the Gallic and Civil wars.

CAMDEN, WILLIAM: Born 1551; died 1623. Antiquary. Published his 'Britannia' 1586–1607.

COLONNA, EGIDIO: Died 1316. Archbishop of Bourges 1294, after having been tutor to Philip the Fair of France. His best known works are 'Quæstiones Metaphysicales' and 'De Regimine Principum'; the latter was written about 1280. Colonna gives a description of the siege engines of his time.

DANIEL, PÈRE GABRIEL: Historian. Born 1649; died 1728.

DIODORUS (The Sicilian): Historian. Lived under Julius and Augustus Cæsar (Augustus died A.D. 14). The best modern edition is that edited by L. Dindorf, 1828.

FABRETTI, RAFFAEL: Antiquary. Born 1618; died 1700.

FROISSART, JEAN: French chronicler. Born about 1337; died 1410. His Chronicles printed about 1500. Translated into English by Lord Berners, and published 1523–1525.

GROSE, FRANCIS: Military historian and antiquary. Born about 1731; died 1791. Published 'Military Antiquities' 1786–1788.

HERON OF ALEXANDRIA: Mechanician. Lived B.C. 284–221. Bernardino Baldi edited his work on arrows and siege engines, 1616 (*vide* Thévenot).

ISIDORUS, BISHOP OF SEVILLE: Historian. Died 636.

JOSEPHUS, FLAVIUS: Jewish historian. Born A.D. 37; died about the year 100. Wrote the 'History of the Jewish Wars' and also 'Jewish Antiquities.' Josephus, acting as commander of the besieged, bravely defended Jotapata, A.D. 67, against the Roman general Vespasian. He was also present with the Roman army during the siege of Jerusalem by Titus, A.D. 70.

LEONARDO DA VINCI: Italian painter. Born 1445; died 1520. In the immense volume of sketches and MSS. by this famous artist, which is preserved at Milan and entitled 'Il Codice Atlantico,' there are several drawings of siege engines.

LIPSIUS, JUSTUS: Historian. Born 1547; died 1606.

MÉZERAY, FRANÇOIS E. DE: French historian. Born 1610; died 1683. Published 'Histoire de France,' 1643–1651.

NAPOLEON III.: 'Etudes sur l'artillerie,' compiled by order of the Emperor and containing many drawings of the full-sized models of siege engines made by his orders, with interesting and scientific criticism of their power and effect.

PHILO OF BYZANTIUM: A writer on and inventor of warlike and other engines. Lived shortly after the time of Archimedes (Archimedes died 212 B.C.): was a contemporary of Ctesibius, who lived in the reign of Ptolemy Physcon, B.C. 170–117 (*vide* Thévenot).

PLUTARCH: Biographer and historian. Time of birth and death unknown. He was a young man in A.D. 66.

POLYBIUS: Military historian. Born about B.C. 204. His History commences B.C. 220 and concludes B.C. 146. The most

interesting edition is the one translated into French by Vincent Thuillier with a commentary by de Folard, 1727–1730.

PROCOPIUS: Byzantine historian. Born about 500; died 565. The best edition is that of L. Dindorf, 1833–1838.

RAMELLI, AGOSTINO: Italian engineer. Born about 1531; died 1590. Published a work on projectile and other engines, 1588.

TACITUS, CORNELIUS: Roman historian. Born about A.D. 61.

THÉVENOT, MELCHISEDECH, 1620–1692: Edited a book called ‘*Mathematici Veteres*,’ containing several treatises on the siege operations of the ancients, including the construction and management of their projectile engines. In this book are to be found the writings on the subject of military engines that were compiled by Athenæus, Apollodorus, Biton, Heron and Philo. Thévenot was King’s librarian to Louis XIV. After his death the manuscript of ‘*Mathematici Veteres*,’ or ‘*The Ancient Mathematicians*,’ was revised and published by La Hire in 1693. The book was again edited by Boivin, an official in the King’s library, who lived 1663–1726. The treatises contained in Thévenot were finally re-edited and published by C. Wescher, Paris, 1869.

VALTURIUS, ROBERTUS: Military author. Living at the end of the fifteenth century. His book ‘*De Re Militari*’ first printed at Verona, 1472.

VEGETIUS, FLAVIUS RENATUS: Roman military writer. Flourished in the time of the Emperor Valentinian II., 375–392. The best edition is that of Schwebel, 1767.

VIOLLET-LE-DUC: French military historian. Published his ‘*Dictionnaire raisonné de l’Architecture*,’ 1861.

VITRUVIUS POLLIO: Architect and military engineer and inspector of military engines under the Emperor Augustus. Born between B.C. 85 and 75. His tenth book treats of siege engines. Translated into French with commentary by Perrault, 1673. The most interesting editions of Vitruvius are those containing the

commentary on siege engines by Philander. The best of these is dated 1649.

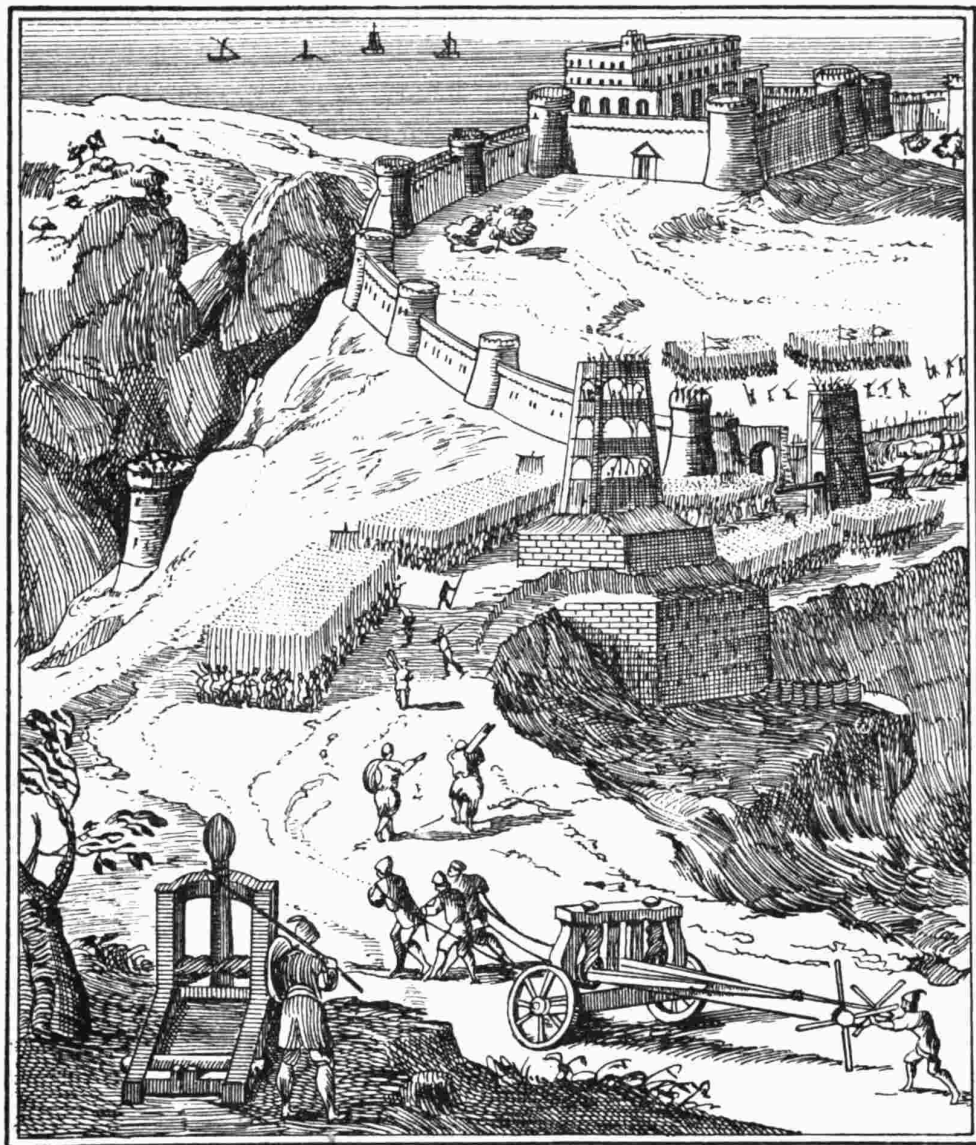


FIG. 21.—THE CAPTURE OF A FORTRESS.

Criticism.—A fortification being entered by the besiegers, who have made a breach in the outside wall with a battering ram.

A catapult is in the left corner of the picture, and four men are taking a balista up the approach to the gateway.

From Polybius. Edition 1727.

**A TREATISE ON
THE CONSTRUCTION, POWER AND
MANAGEMENT
OF TURKISH AND OTHER ORIENTAL
BOWS
OF MEDIAEVAL AND LATER TIMES**

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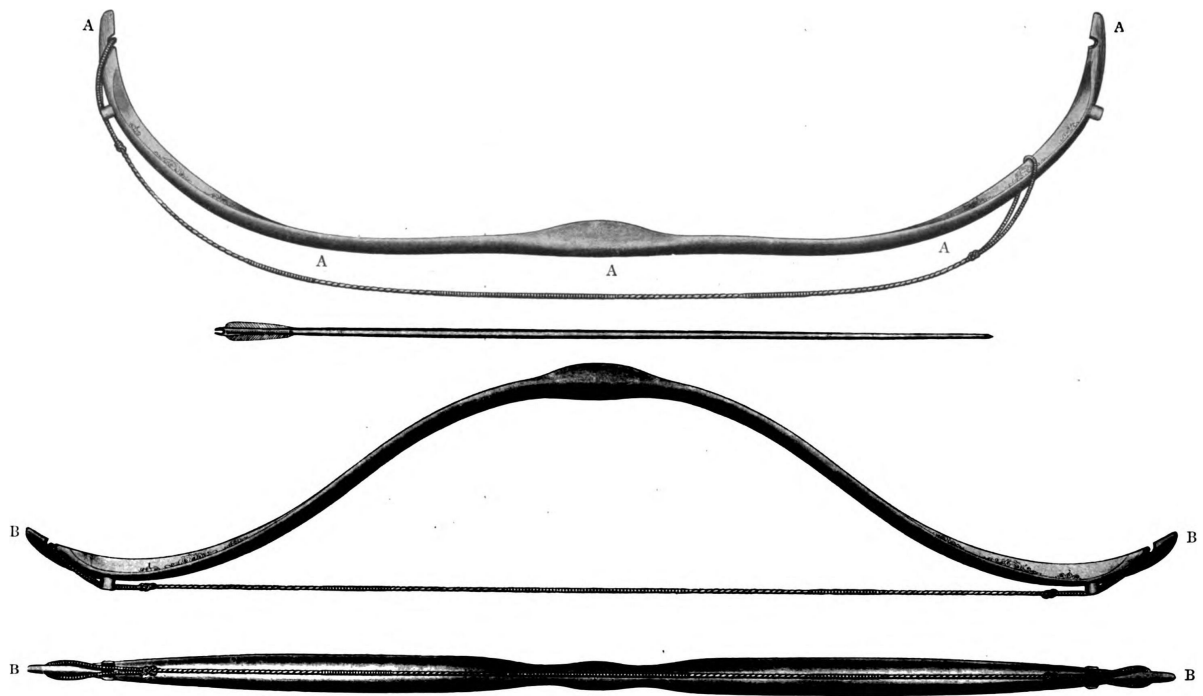


FIG. 1.—TURKISH REFLEX COMPOSITE BOW, UNSTRUNG AND STRUNG, AND ITS FLIGHT ARROW.

PART I

THE TURKISH BOW—CONSTRUCTION AND DIMENSIONS

LENGTH of bow, measured, before it is strung, from end to end along its outer curve with a tape, 3 ft. 9 in. (AAAAA [fig. 1](#), opposite page).

Span of bow, measured between its ends when strung, 3 ft. 2 in. (BB [fig. 1](#).)

Length of bow-string, 2 ft. 11 in.

Greatest width of each arm of bow, $1\frac{1}{8}$ in.

Thickness of each arm, at a distance of 6 in. from the centre of the handle of the bow, $\frac{1}{2}$ in. ⁴³

⁴³ In the very powerful bows, such as the one shown in [Fig. 15](#), p. 121, the thickness at these parts is from $\frac{5}{8}$ to $\frac{3}{4}$ in.

Circumference of each arm, at a distance of 6 in. from the centre of the handle of the bow, 3 in.

(The arms of the Persian, Indian, and Chinese composite bows have a width of from $1\frac{1}{2}$ to 2 in.; and though the span of these bows, when strung, is from 4 to 5 ft. and more, they do not shoot a light arrow nearly so far as the shorter, narrower, and in proportion far stronger and more elastic Turkish ones.)

The strength of the bow, or the weight that would be required on the centre of the bow-string to pull it down from the bow to the full length of the arrow, is 118 lbs. (This is without taking into account the additional two

or three inches the point of the arrow should be drawn within the bow along the horn groove.)

Weight of bow, avoirdupois, 12½ oz.

Though I have carefully examined over fifty of these small Turkish bows, I have never seen one that exceeded 1¼ in. in width at its widest part, or if measured with a tape along its outer curve, when unstrung (AAAAA, [fig. 1](#)), was over 3 ft. 10 in. in length. Bows that are 4 or 5 in. longer than the dimensions here given are invariably of Persian or Indian manufacture, and are very inferior in the elasticity that is requisite for long-distance shooting, though in decoration and construction they often closely resemble Turkish bows.

The bow is chiefly constructed of very flexible horn and sinew. These materials were softened by heat and water and then longitudinally glued to a slight lath of wood varying from ⅛ to ¼ in. in thickness (except where it formed the handle of the bow), and from ½ to 1 in. in width.

This strip of wood formed the core or mould of the bow, and extended at each of its ends for 3 in. beyond the strips of horn and sinew that were fixed on its opposite sides, and which slightly overlapped it. ([Fig. 2](#), p. 105.) The projecting ends of the wooden strip were enlarged so as to form the solid extremities of the bow in which the nocks for the bow-string were cut. (CC [fig. 3](#), p. 106.)

The two curved horn strips, which in part comprised the arms of the bow (on its inside face when it was bent), were cut from the horn of a buffalo or an antelope, and average about ¼ in. in thickness.

The thicker ends of these pieces meet at the middle of the handle of the bow and their tapered ends extend to within 3 in. of its wooden points. (EE [fig. 3](#), p. 106.)

The sinew that represents the back of the bow is from the great neck tendon of an ox or stag. This was probably shredded longitudinally, and, after being soaked in elastic glue, compressed into a long flat strip about ¼ in. thick, which was first moulded in a pliable state to the wooden core and then glued to it. It thus formed the back of the bow when it was bent. (DDD [fig. 3](#), p. 106.)

The bark of the cherry-tree, or thin leather or skin, was next glued over the sinew to preserve it from injury and damp. The horn parts, or inner face of the bow when it was strung, were not covered with bark or skin, a feature of the Turkish bow that, together with its small size, distinguishes it from the bows of India and other Oriental countries.⁴⁴

⁴⁴ Though the horn strips which form the belly, or inner surface when it is strung, of a Chinese or a Tartar bow, are neither covered nor decorated, the great size of these weapons easily distinguishes them from those of Turkish manufacture. ([Figs. 13, 14](#), pp. 116, 117.)

In the best Turkish bows this outer coating of bark, leather or skin was lacquered a brilliant crimson and elaborately decorated with gold tracery, the date of the bow being always placed at one of its ends and the name of its maker at the other.

The horn and sinew (the materials which really form the bow and give it its power and elasticity) may be likened to a tube, the small centre of which is filled with wood. (Sections, [fig. 2](#), opposite page.)

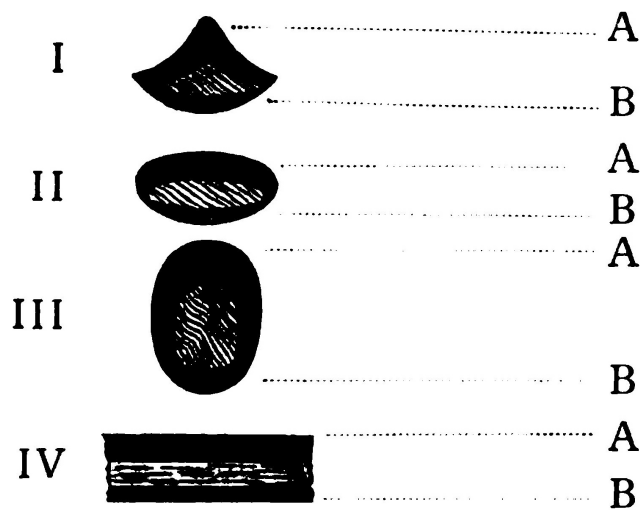


FIG. 2.—SECTIONS OF A TURKISH BOW
Half full size.

I. Section of bow at 6 in. from one of its ends.

II. Section of bow at half-way between the centre of its handle and one of its ends.

III. Section of bow at the centre of its handle, which is here thickly covered with sinew.

IV. Longitudinal section of bow at half-way between the centre of its handle and one of its ends.

Light shading, AAAA. The compressed sinew forming the back of the bow when it is strung.

Dark shading, BBBB. The horn forming the inner surface of the bow when it is strung.

Lined centres. The thin lath of wood to which the horn and sinew parts of the bow are moulded and fixed.

The thin wooden lath, in places only $\frac{1}{8}$ in. thick, bestowed no strength on the bow, as it was merely its heart or core to which the two curved strips of horn and the long band of sinew were glued. ([Fig. 3](#), p. 106.)

As it would have been very difficult and tedious to shape so fragile a lath in one length to suit the outline of the finished bow, this lath was always made in three pieces, which were fitted together at their joints and then secured with glue. ([Fig. 3.](#))

The middle piece formed the core of the handle of the bow and the other pieces the core of its limbs. ([Fig. 3.](#))

The extremities of the two outer pieces of the core were enlarged to form the strong projecting points of the bow in which the nocks for the bow-string were cut. (CC [fig. 3.](#))

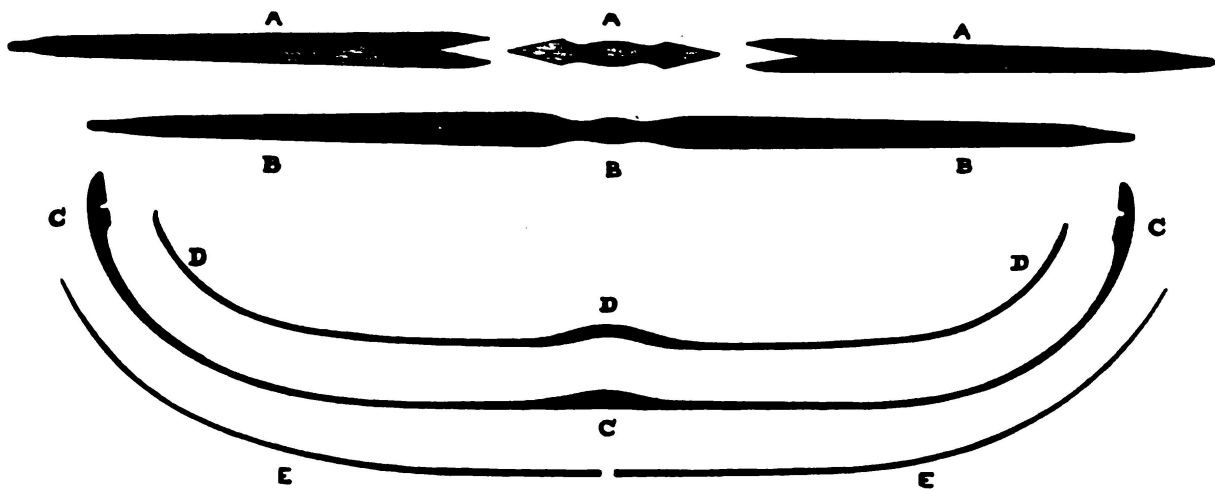


FIG. 3.—LONGITUDINAL PLANS OF THE PARTS OF A TURKISH BOW.

AAA. The three pieces of thin wood that formed the core of the bow. Surface view. (The two outer lengths of the core were steamed into a curve as shown in CCC.)

BBB. The pieces glued together. Surface view.

CCC. The pieces glued together. Side view.

DDD. The strip of sinew that was glued to the core, and which formed the back or outer surface of the bow when it was reversed and strung.

EE. The two strips of naturally curved horn that were glued to the core, and which formed the belly or inner surface of the bow when it was

reversed and strung.

PART II

THE BOW-STRING

THE main part of the bow-string was composed of a skein of about sixty lengths of strong silk and was ingeniously knotted at each of its ends to a separate loop, formed of hard and closely twisted sinew. A loop and its knot is shown in [fig. 4](#), opposite page.

These loops could not fray or cut, as would occur if they were made of silk, and they fit into the nocks of the bow. The loops rest, when the bow is strung, upon small ivory bridges ([fig. 1](#), p. 102) which are hollowed out to receive them, and which, in this way, retain the bow-string in its place. Though these little bridges are not always present on Turkish bows, they are invariably to be found on those of Persian, Indian or Chinese construction, their greater length requiring the assistance of bridges to keep their bow-strings in a correct position.

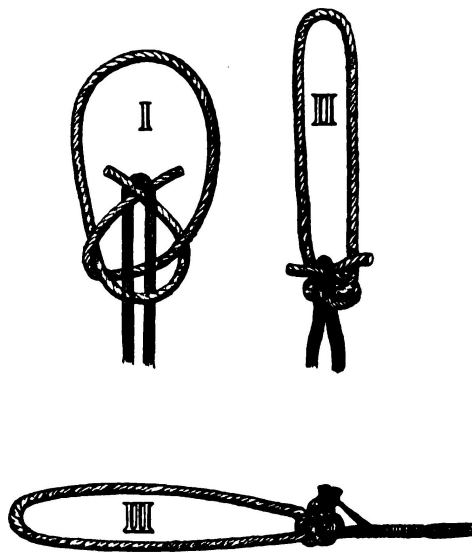


FIG. 4.—ONE OF THE LOOPS OF HARD AND CLOSELY TWISTED SINEW WHICH ARE KNOTTED TO EACH END OF THE MIDDLE PART OR SKEIN OF A TURKISH BOW-STRING.

Scale: Half full size.

I. A loop and its knot as first formed on one end of the skein of the bow-string.

II. The loop drawn up, but not tightened.

III. The loop drawn up tight and its loose ends secured.

As shown in III, the projecting ends of the length of sinew which forms the loop are cut off to within a third of an inch of the knot. They are singed at their extremities, so as to form small burrs which prevent the short length of strong silk, which lashes them together, from slipping off.

The ends of this last small lashing are placed beneath the wrapping of silk to be seen on the skein near the knot in III.

In this way the knot of the loop is rigidly secured against any chance of drawing when the bow is in use.

(The bow-strings of all Oriental bows, with the exception of the Tartar and Chinese, were made as above described.)

PART III

THE ARROW

LENGTH of arrow, $25\frac{1}{2}$ in. to $25\frac{3}{4}$ in.

Weight of arrow, avoirdupois, 7 drs., or equal to the weight of two shillings and a sixpence.

The balance of the arrow is at 12 in. from the end of its nock.

Shape of arrow, 'barrelled,' and much tapered from its balancing-point to its ends: its sharp ivory point being only $\frac{1}{8}$ in. in diameter (where it is fitted to the shaft) and $\frac{1}{4}$ in. in length. The part of the shaft to which the feathers are attached is $\frac{3}{16}$ in. in diameter, and the centre of the shaft $\frac{5}{16}$ in.

Though I have carefully measured and weighed about two hundred eighteenth-century Turkish flight arrows, I have scarce found a half-dozen that were $\frac{1}{8}$ in. more or less than from $25\frac{1}{2}$ in. to $25\frac{3}{4}$ in. in length, or that varied by even as little as $\frac{1}{2}$ dr. from 7 dr. in weight. In regard to their balancing-point these arrows are equally exact, as this part is invariably from $11\frac{1}{2}$ in. to $12\frac{1}{2}$ in. from the nock.

It is evident that the old Turkish flight arrow was accurately made to a standard pattern that experience showed was the most successful one for long-distance shooting.

The light and elegantly shaped wooden nock of an old Turkish arrow ([fig. 5](#)) is quite unlike the clumsy horn nock of the modern European one.

The latter cannot withstand the recoil of the Turkish bow and soon splits apart, though in the thousands of times I have discharged Turkish arrows I have never known one to split at the nock.

It will be noticed that the shape of the Turkish nock—with its narrow entrance that springs apart to admit the bow-string and then closes again—

enabled an archer, even on horseback, to carry an arrow ready for use on the string of his bow.

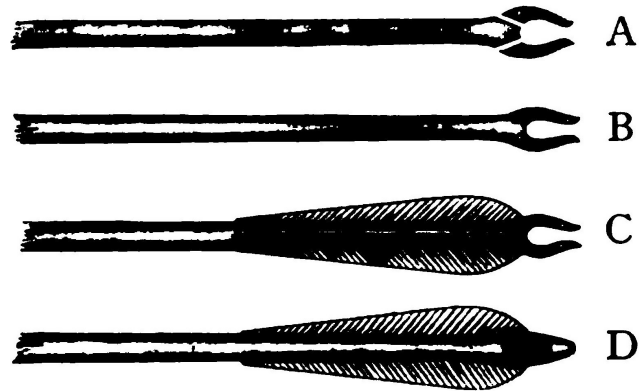


FIG. 5.—THE CONSTRUCTION OF THE
NOCK OF A TURKISH ARROW.

Scale: Half full size.

A. The butt end of the arrow, with the projecting wooden halves of the nock shaped and ready to be glued to the shaft.

B. The halves of the nock glued to the shaft.

C, D. The feathers glued to the shaft.

The feathers (3) of a Turkish flight arrow, though stiff, are as thin as paper, and are $2\frac{1}{2}$ in. long and $\frac{1}{4}$ in. high near the nock. They were often made of parchment.

The dark band of shading to be seen round the nock in C and D is a wrapping of fine thread-like sinew. This sinew, after being soaked in hot glue, was wound to a thickness of about $\frac{1}{32}$ in. all over the nock and it thus held the halves of the latter securely to the shaft.

When dry, the wrapping of sinew was cut out where it crossed the opening for the bow-string. It nevertheless gave a great increase of strength to the thin projecting halves of the nock, as it covered them on their outer surfaces with a sheathing that was very tough and elastic, and as smooth as glass to the touch. This wrapping was, of course, applied before the feathers were glued on.

So careful were the Turks in the construction of these arrows, that even the halves of their nocks were made from wood with a natural curve to suit the finished outline. It is possible, of course, they would not otherwise have withstood the violent shock of the released bow-string. It may be said that every inch in length of a Turkish bow or arrow was named in a manner that could be recognised or referred to. In a general way the parts of an arrow were known as follows:—

The enlarged centre	the stomach.
From the centre to the point	the trowser.
From the centre to the nock	the neck.

PART IV

THE METHOD OF STRINGING A TURKISH, PERSIAN OR INDIAN BOW.

IN these days no person I have ever heard of can string a strong Turkish bow—diminutive as this weapon is—without much personal assistance, or else by mechanical means, yet formerly the Turkish archer unaided could do so with ease.

This he achieved by a combination of leg and manual power. ([Figs. 6 and 7](#), p. 110.)

With the longer reflex bows, the Chinese for instance, this operation is comparatively easy, as the hand can reach one end of the bow and draw it inwards for the loop of the bow-string to be slipped into the nock.

The Turkish bow, being so short, necessitates a great effort of strength on the part of the archer to bend it between his legs and, at the same time, stoop down to fit the bow-string. From constant practice, the Turk of former days knew exactly how and when to apply the muscular force of leg and arm necessary to string his bow—a performance that no modern archer could accomplish with a bow of any strength.

Leg and manual force combined is the only possible method of stringing a strong reflex bow, unless mechanical power is utilised: it was the hereditary custom of the Orientals. In the operation, there is always the risk of twisting the limbs of the bow, from a lack of the great strength of wrist required to hold them straight during the stringing. If the limbs of the bow are given the slightest lateral twist as they are being bent, the horn parts are certain to splinter, and the bow is then useless and damaged beyond repair.^{[45](#)}

⁴⁵ The only safe method for a modern archer to adopt in order to string a powerful reflex bow is to use strong upright pegs, the size of tent pegs, inserted in smooth ground or in holes in a board, the bow resting during the process flat along the ground or board. Insert one peg against the inner face of the handle of the bow and then pull the ends of the bow back by degrees, placing a peg behind each of its ends as you do so to retain them in their acquired positions. The outer pegs can be shifted towards you as the bow is gradually bent, first at its one end and then at its other one. Finally, when the bow is fully bent, the bow-string can be fitted across it from nock to nock and the pegs removed. To unstring the bow, grasp its extremities and, with the palms of the hands uppermost, bend it slightly across the knee, at the same time shifting with the thumb one of the loops of the bow-string out of its nock.

The difficulty of reversing and stringing a very stiff bow with such a reflex curve that its ends nearly meet before it is bent may be imagined.

De Busbecq tells us that some of the Turkish bows were so strong that if a coin was placed under the bow-string at one end of the bow, as it was being strung, no one but a trained archer could bend the bow sufficiently to set free the coin so that it fell to the ground.



FIG. 6.

[Fig. 6](#) shows an Oriental reflex bow being gradually reversed preparatory to fitting on its bow-string.

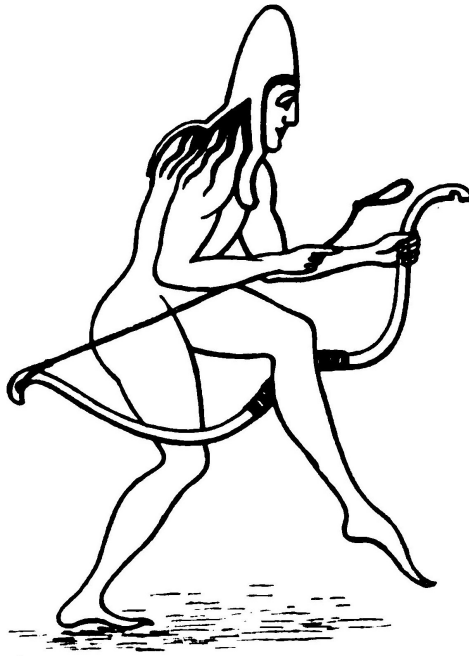


FIG. 7.

[Fig. 7](#) shows a similar bow when reversed sufficiently to fit its bow-string.

Though this illustration is from an ancient Greek vase, it will be noticed that in it the power of the leg and arm is applied in precisely the same way as in the more modern example given.

PART V

THE HORN GROOVE

THE thin horn groove which the Turk wore on the thumb of his left hand when flight-shooting is shown in [fig. 8](#).

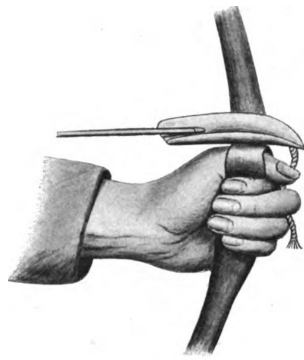


FIG. 8.—THE HORN GROOVE.

The bow is shown fully bent and ready for release, the point of the arrow being drawn back for a couple of inches inside the bow.

This ingenious contrivance enabled the archer to draw the point of his arrow from 2 to 3 in. within the inner surface of his bent bow. He was thus able to shoot a short and light arrow, that would fly much farther than the considerably longer and heavier one he would have had to use if he had shot in the ordinary manner without the grooved horn.

The groove in the horn guides the arrow in safety past the side of the bow, when the bow-string is released by the archer.

The Turk, in fact, shot a short and light arrow from a very powerful bow, which he bent to the same extent as if he used an arrow 3 in. longer, with its proportionately increased size, weight, and frictional surface to retard its flight.

In the former case it will easily be understood that a much longer range could be achieved than in the latter.

Of this increase in length of flight conferred by the use of the grooved horn, the following experiment is conclusive evidence.

I lately shot from a Turkish bow twelve arrows, each arrow being three-quarters of an ounce in weight and 28½ in. in length.

These twelve arrows were individually drawn to the head and the distance they reached averaged 275 yards.

I then reduced the same arrows to a length of 25½ in. each, and to a weight of half an ounce each.

They were now shot from the same bow, over the same range and under the same conditions of weather, but their points were drawn 2½ in. within the bow along a grooved horn. The distance they then travelled averaged 360 yards.

The Turk, as was the custom of Orientals, shot his arrow from the right-hand side of his bow, as shown in [fig. 8](#), p. 111.⁴⁶

⁴⁶ To discharge the arrow from the left-hand side of the bow, as is the custom in all European archery, the leather ring and the grooved horn will have to be fitted to the first joint of the forefinger.

The bow is here represented as fully bent, the point of the arrow being drawn back along the groove of the horn for a couple of inches within the bow.

The horn is attached to the thumb by a small leathern collar.

A short plaited cord of soft silk is suspended from the fore-end of the horn and is gripped between the fingers of the archer as he holds the bow.

This cord enables the archer to keep the horn in a level position on his hand. It is fixed to a small strip of leather which is glued beneath the horn.

The horn is usually of tortoiseshell, very highly polished. It is from 5 to 6 in. long, 1 in. wide, $\frac{1}{4}$ in. deep inside and $\frac{1}{16}$ in. thick.

It is slightly sloped from its centre of length to each of its ends, so that when the arrow is projected it touches the hard and smooth surface of the horn very lightly, and with, therefore, the least possible friction to retard its flight.

As the horn groove is only one-sixteenth of an inch thick, the arrow, as it is drawn back or shot forward, may be said to fit close against the side of the bow.

PART VI

THE THUMB-RING.

THE Turk pulled his bow-string with a ring of ivory, or of other hard material, fitted on his right thumb. ([Fig 9](#), p. 113.) Its manipulation is shown on [p. 114](#).

It might be supposed that the strain of the bow-string on the ivory ring would cause the edges of the latter to injure the flesh and sinews of the thumb; this is not, however, the case in the least.

I find I can bend a strong bow much easier and draw it a great deal farther with the Turkish thumb-ring than I can with the ordinary European finger-grip.

The release to the bow-string which is bestowed by the small and smooth point [in Turkish “lip”] of the thumb-ring, is as quick and clean as the snap of a gunlock when a trigger is pulled, and very different in feeling and effect from the comparatively slow and dragging action that occurs when the release takes place in the modern way from the leather-covered tips of three fingers.

The range of a flight arrow when shot from a bow by means of a thumb-ring is always much beyond that of an arrow shot with the three fingers in the usual manner.

With the thumb-ring the feathers of an arrow can be placed close to its nock, as the usual space of about 1½ in. need not be left on the shaft at the butt-end lest the fingers holding the bow-string should crush the feathers of the arrow—a precaution that is necessary in all European archery.

There is no doubt that the closer to the nock the feathers of an arrow can be fixed, the farther and steadier it will travel.

The handle of an English bow, or of any other bow that is loosed with the fingers, is placed below its centre so that the arrow can be fitted to the middle of the bow-string, a point which is just above the hand of the archer as he grasps the bow.

A bow held below its centre can never be pulled really true, the limb below the handle being shorter than the one above it.

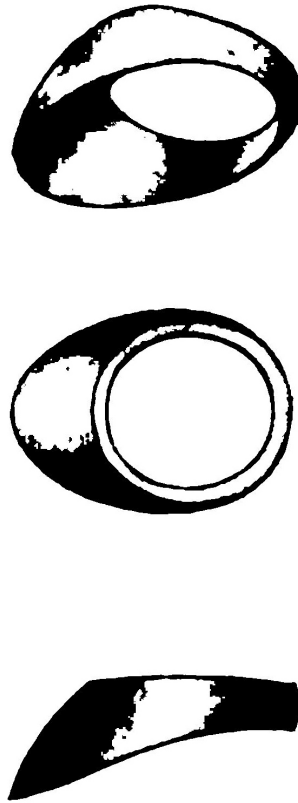


FIG. 9.—

THE TURKISH THUMB-
RING. (Scale, half full
size.)

In a Turkish bow the handle is in its exact centre of length, and the projecting point, or lip, of the thumb-ring engages the bow-string close to its centre.

For these reasons the bow is equally strained, each of its limbs doing its proper share of work in driving the arrow, an advantage that is very noticeable in flight-shooting, and would probably also be at the target. In the method of loosing used in modern times the bow-string lies across the three middle fingers, its outline, where the arrow is nocked on the string, taking the form of two angles connected by a straight line $2\frac{1}{2}$ to 3 in. in length.

With the thumb-ring the bow-string is drawn back to one sharp angle close to the apex of which the nock of the arrow is fitted, so that every part of the string is utilised in driving the arrow. ([Fig. 12](#), p. 114.)

The ease with which a strong bow can be drawn with the thumb-ring, and the entire absence of any unpleasant strain on the thumb, is remarkable. This proves how effective the Oriental style of loosing a bow-string was, compared with the one now practised by European archers.

The ring was usually of ivory, its edges being round and smooth where they came in contact with the skin of the thumb.

A covering of soft leather was sometimes glued all over the sloping outer surface of the projecting lip of the ring.

The leather assisted the archer to hold the ring firmly with his forefinger, so that it could not slip under the strain of pulling back the bow-string. The projecting lip of the ring bestowed the leverage which enabled the archer to draw the bow-string of a powerful bow.

Thumb-rings of silver or of agate were often permanently worn by Turkish archers of position, both for ornament and for use.

These rings were finely polished and frequently inlaid with gold.

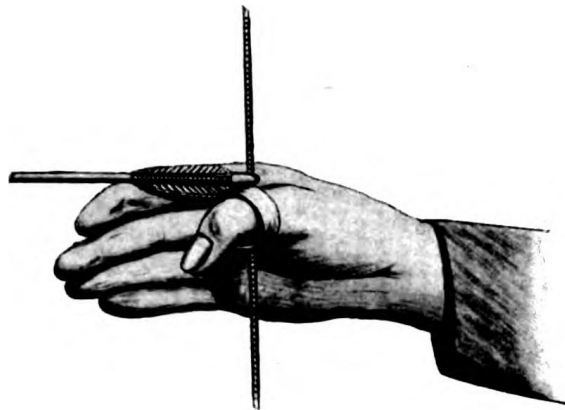


FIG. 10.

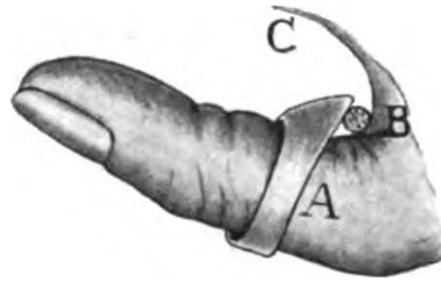


FIG. 11.

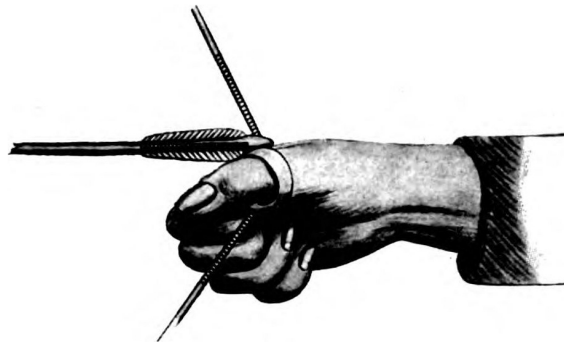


FIG. 12.

THE TURKISH THUMB-RING AND ITS MANIPULATION.

[Fig. 10.](#) The position of the hand when the arrow is first fitted to the bow-string, the latter being hitched behind the lip of the thumb-ring. The nock of the arrow should be close against the lip of the ring, and hence within about an eighth of an inch of the angle formed in the bow-string when it is fully drawn, as shown in [fig. 12](#).

[Fig. 11.](#) View of the thumb, with the ring, A, in position preparatory to closing the forefinger and thumb.

[B. Section of the bow-string as hitched behind the projecting lip of the ring.

C. The base of the forefinger, or the part of it which presses tightly over the sloping surface of the lip of the ring, in front of the bow-string,

when the bow is being bent.]

[Fig. 12.](#) The base of the forefinger pressed against the ring, the hand closed, and the bow-string and arrow being drawn back by the thumb-ring.

It should be noted that no part of the hand is utilised in holding the ring and in drawing the bow-string, except the thumb and the base of the forefinger.

When the pressure of the forefinger is taken off the ring (by separating this finger and the thumb) the bow-string instantly pulls the lip of the ring slightly forward, and at the same moment slips off it with a sharp ‘click.’

The archers of other Oriental nations besides the Turks employed thumb-rings of various shapes and dimensions to suit the construction of their bows, bow-strings and arrows. All thumb-rings were, however, more or less similar, and were all used in the manner I have described.

It is, indeed, impossible to shoot an arrow by means of a thumb-ring except as I have shown, and as a very short practical trial will prove.

If the ring is applied in any other way it either flies off the hand when the bow-string is released; the thumb is injured; or the bow-string escapes from its hold when only partially drawn.

* * * * *

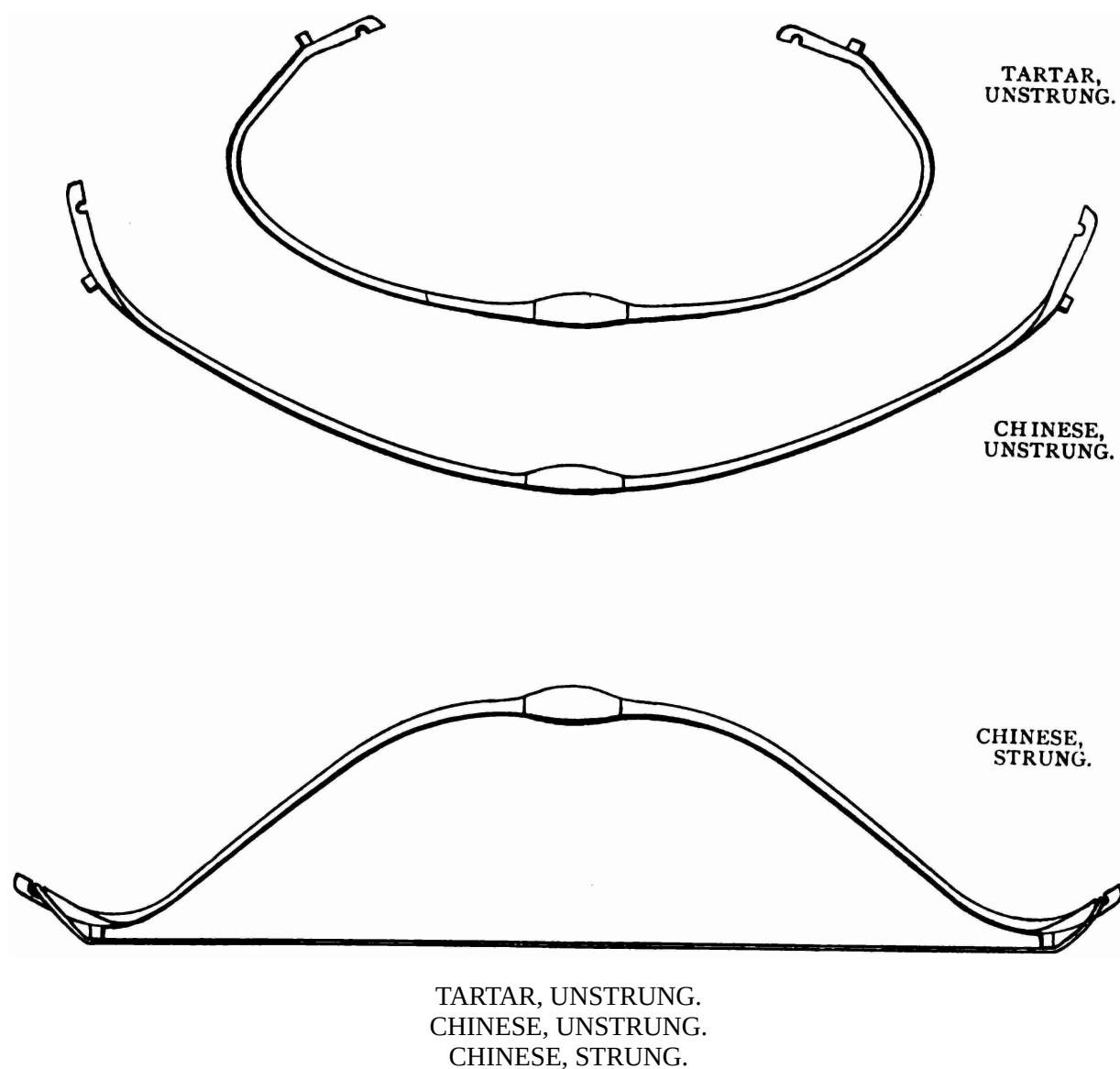
In one of the Turkish manuals on Archery translated by Baron Purgstall, many illustrations are given of the construction of the Turkish composite bow, but, unfortunately, minor details are omitted, though doubtless they were common knowledge when the Ottoman author wrote.

Without these details the correct formation of the bow cannot be ascertained. The chief omissions are (1) the composition of the very strong and elastic glue with which the parts of the bow were so securely joined, (2) the treatment of the flexible sinew which formed the back of the bow—whether, for instance, it was glued on in short shredded lengths or was attached in one solid strip.

All we know is that the sinew was taken from the *Ligamentum Colli* of an ox or stag, a very powerful and elastic tendon which contracts or expands as the animal raises or lowers its head to feed or drink.

When the sinew which comprises the back, or outside when it is strung, of a Turkish bow—however old it be—is dissolved in hot water, it disintegrates into hundreds of short pieces of from 2 to 3 in. long and about $\frac{1}{8}$ in. in diameter, each as ductile as indiarubber and almost unbreakable by hand.

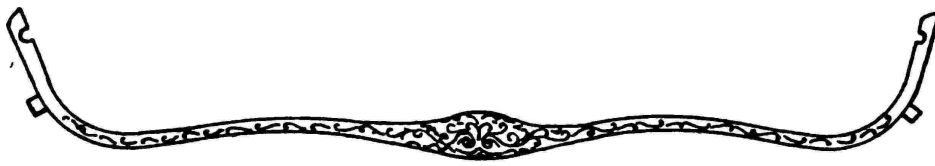
The component parts of a Turkish bow, consisting of a thin strip of horn, one of wood and another of sinew ([fig. 3](#), p. 106), are so pliable when separated that they can almost be coiled round the fingers, though if the same pieces are glued together they form a bow of unrivalled strength and elasticity.



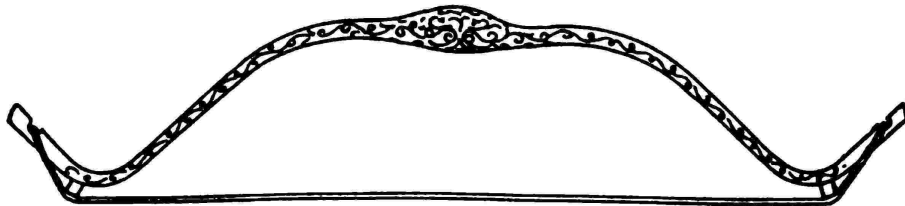
Scale: One inch = one foot.

FIG. 13.

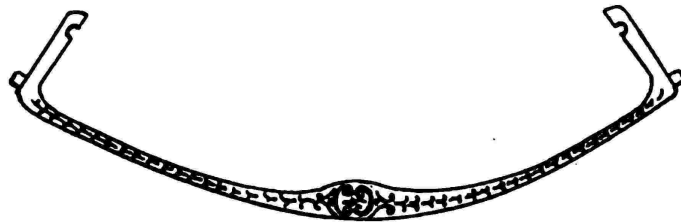
[Figs. 13, 14.](#) THE COMPARATIVE DIMENSIONS OF THE REFLEX COMPOSITE BOWS OF VARIOUS NATIONS.—The structure of all these bows is similar in that they are composed of sinew, wood and horn, *i.e.* sinew on the back of the bow, naturally curved horn on its inner face, and a thin core of wood between the horn and sinew.



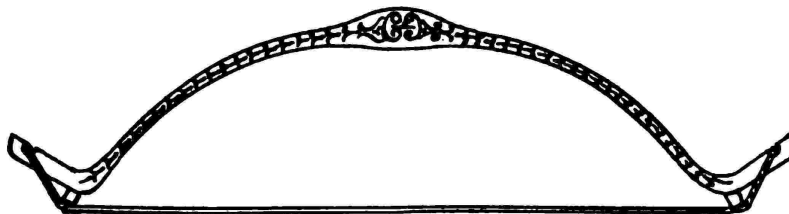
PERSIAN,
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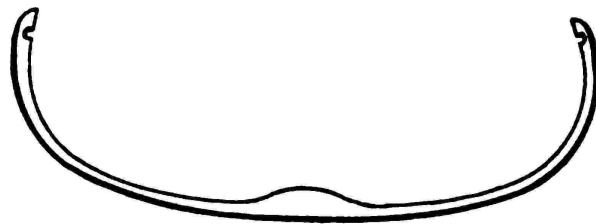
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STRUNG.



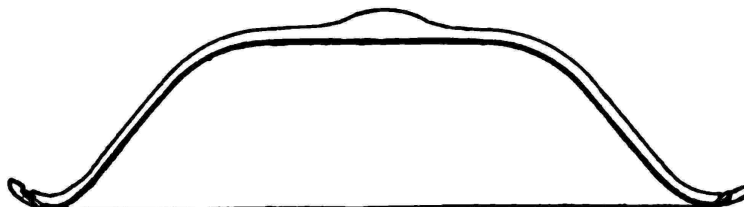
INDIAN,
UNSTRUNG.



INDIAN,
STRUNG.



TURKISH,
UNSTRUNG.



TURKISH,
STRUNG.

PERSIAN, UNSTRUNG.
PERSIAN, STRUNG.
INDIAN, UNSTRUNG.
INDIAN, STRUNG.
TURKISH, UNSTRUNG.
TURKISH, STRUNG.

Scale: One inch = one foot.

FIG. 14.

Though the range of the Turkish bow—whether with a flighting or with a war arrow—far exceeds that of the other bows depicted, yet the Persian and Indian weapons are capable of shooting to a long distance, certainly much farther than any European longbow.

The great Chinese or Tartar bow requires a very long arrow, which from its length is, of necessity, a heavy one with a thick shaft. It cannot be propelled, as a result, farther than from 250 to 260 yards. One distinctive feature of Chinese, Tartar, Persian or Indian bows is the formation of their bow-strings. These are invariably from $\frac{1}{4}$ to $\frac{5}{16}$ in. in thickness, and are always closely wrapped round, from end to end, with soft cord or coloured silk of about the substance of worsted.

The Turkish bow-string is $\frac{1}{8}$ in. thick, and is merely served round with fine silk for 3 in. at its centre of length, with three or four shorter lashings at intermediate points.

THE LENGTHS OF THE ARROWS FORMERLY USED IN WARFARE WITH THE BOWS GIVEN
IN FIGS. 13 AND 14.

Chinese or Tartar bow	3 ft.
Persian	2 ft. 8 in.
Indian	2 ft. 6 in.
Turkish ⁴⁷	2 ft. 4½ in.

⁴⁷ The long Turkish war arrow was drawn to the head as in an ordinary bow. The grooved horn was only used with the short and light flight-arrow.

PART VII

THE RANGE OF THE TURKISH BOW



THE AUTHOR SHOOTING WITH A
TURKISH BOW.⁴⁸

IN 1795 Mahmoud Effendi, Secretary to the Turkish Ambassador in London, shot a 25½-in. flight arrow 480 yards. The bow he used is similar to the one shown in [fig. 11](#), p. 112, and is now preserved in the Hall of the Royal Toxophilite Society, Regent's Park.

Mahmoud Effendi accomplished this feat—which was carefully verified at the time—in the presence of a number of well-known members

of the Toxophilite Society of the day, including Mr. T. Waring, the author of a work on Archery.

Joseph Strutt, the historian, was also a spectator, and describes the incident in his book entitled 'The Sports and Pastimes of the People of England.'

⁴⁸ There are many country residences in England at which the author has made very long shots with a bow and arrow, and where trees have been planted to mark the distances. Among others: Glynllivon Park, Carnarvon; Broomhead Hall, Sheffield; Onslow Hall, Shrewsbury; Norton Priory, Runcorn; and Harpton Court, New Radnor, may be named.

It is beyond question that in the seventeenth and eighteenth centuries, with bows precisely similar to the one shown in [Fig. 1](#), but of much greater power, flight arrows were shot from 600 to 800 yards by certain famous Turkish archers.

The achievements of these celebrated bowmen were engraved on marble columns erected at the ancient archery ground near Constantinople, and these records are still in existence ([p. 125](#)).

The only trustworthy evidence of unusual ranges attained with the English longbow is as follows:

1798.	Mr. Troward	340	yards.
1856.	Mr. Horace Ford	308	„
1881.	Mr. C. J. Longman	286	„
1891.	Mr. L. W. Maxon	290	„
1897.	Major Joseph Straker	310	„

It is not probable that the English bowmen of mediæval days were able to shoot the arrows they used in warfare farther than from 230 to 250 yards. Nor is it likely that they could send flight arrows to longer ranges than those given above, as heavy yew bows, strong as they may have been, were

unsuitable for the purpose.⁴⁹ It was from their great elasticity, as much as from their strength, that composite bows derived their wonderful power.

⁴⁹ In King Henry IV., Second Part, Act III., Scene 2, Shakespeare makes Shallow exclaim of Double that the latter could shoot a flight arrow from 280 to 290 yards. In the time of Shakespeare (1564–1616) it was, therefore, considered a notable feat to send an arrow to this distance.

When, too, the composite bow was strung, its bow-string was much more taut than was that of any European bow, as the latter was merely bent out of a straight line, whilst the former was bent from a sharp reflex curve, which it was always striving to resume when in use.

Though many nations formerly used composite bows of horn and sinew, no people attained such dexterity in their manipulation, or constructed them of such marvellous power and efficiency, and at the same time so small, elegant and light, as did the Turks.

It should not be supposed, however, that because these bows were so diminutive in size, they were mere playthings for shooting a flight arrow to an immense range. They were powerful weapons of warfare, and, as I have proved in practice, those of only moderate power are capable of sending an iron-shod arrow weighing 5s., or one ounce, to a distance of 280 yards. Bows that could shoot a flight arrow 600 yards, and more, would certainly be able to drive an ounce arrow 360 to 400 yards—or much farther than was possible with the old English longbow and its war shaft.

I have obtained with much difficulty during the last few years about a score of composite bows of Turkish manufacture from various parts of the Ottoman Empire. Not more than three or four of these have, however, proved serviceable, owing to their age, as no bows of the kind have been made for over a hundred years, the art of their construction being long since neglected and lost.

With the bow depicted in [Fig. 1](#), I shot six arrows in succession to ranges exceeding 350 yards, the longest flights being 360, 365 and 367

yards. This public record was established July 7th, 1905, at an archery meeting held at Le Touquet, near Etaples in France. The ground selected for the trial was perfectly level; there was no wind, and the distances were accurately measured by several well-known members of the Royal Toxophilite Society who were present.

With the same bow I have, in private practice, thrice exceeded 415 yards, and on one occasion reached 421 yards.

Though this bow is a powerful one for a modern archer to draw, it is a mere plaything compared with other Turkish bows of the same length, but of far greater strength, which I possess.

Some of the latter are so curved in their unstrung state that their ends nearly meet, and are so stiff, when strung, that I cannot draw them to more than half the length of a 25½-in. arrow. [Fig. 15](#) shows a bow of this kind in my collection.

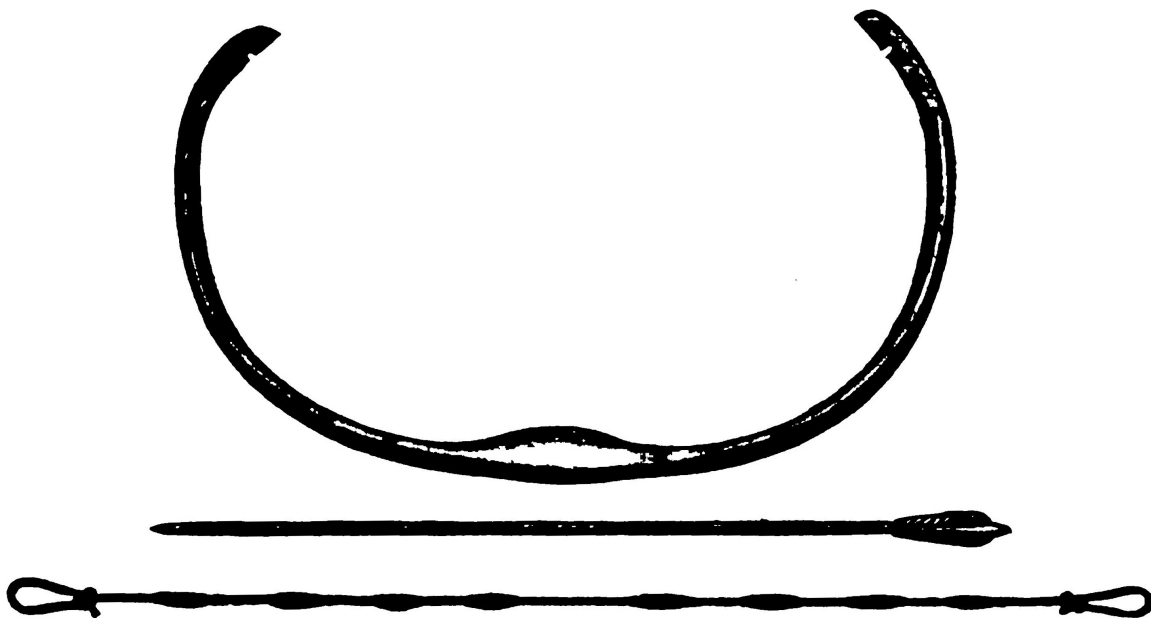


FIG. 15. SKETCH OF A VERY POWERFUL TURKISH BOW WITH ITS ARROW AND BOW-STRING.

Such bows as these require a pull of 150 to 160 lbs. to bend them to their full extent, which quite accounts for the marvellous, but well

authenticated, distances attained in flight-shooting by the muscular Turkish bowmen of bygone days.

Though 367 yards is a short range in comparison with that which the best Turkish archers were formerly capable of obtaining, it is, so far as known, much in excess of the distance any arrow has been shot from a bow since the oft-quoted feat of Mahmoud Effendi in 1795, [p. 119](#).

Full corroboration of the wonderful flight-shooting of the Turks may be found in some treatises on Ottoman archery which have been translated into German by Baron Hammer-Purgstall (Vienna, 1851).

In his directions concerning the selection of suitable bows and arrows for the sport, one of the Turkish authors quoted by Purgstall writes: ‘The thinnest and longest flying arrow has white swan feathers shaped like leaves,^{[50](#)} and this arrow, with a good shot, carries from 1,000 to 1,200 paces.’

⁵⁰ *Anglice*, Balloon feathers.

The orthodox length of a pace is thirty inches, and thus even 1,000 paces, or the lesser range mentioned, would exceed 800 English yards.

Augier Ghislen de Busbecq (1522–1592), a Belgian author and diplomatist, describes the Turkish archery he witnessed when ambassador to the court of Solyman, and the well-nigh incredible distances to which he saw arrows propelled.

Full information to the same effect, with excellent diagrams, may be found in a Latin MS. on Turkish archery by J. Covell, D.D., Chaplain to the Embassy at Constantinople 1670–1676.^{[51](#)}

⁵¹ MSS., B.M., 22911, folio 386.

Another treatise (in Turkish) entitled ‘An Account of some famous Archery Matches at Bagdad (1638–1740),’ dedicated to the Governor of that city by the author, M. Rizai,⁵² may also be consulted, as it gives the exact ranges of the longest-flying arrows.

⁵² Sloane MSS., B.M., 26329, folio 59.

It should be remembered that many years ago flight-shooting was a very popular recreation of the Turks, that every able-bodied man was a practised archer, and that every male child was trained to use a bow from the earliest possible age.

The origin of Turkish and other highly finished composite bows, and the approximate date when they were first used in sport and warfare, it is now impossible to determine. Bows that are undoubtedly of this kind and which are of excellent shape and design, are depicted on some of the most ancient pottery existent, and are also referred to in some of the oldest writings we possess.

In further connection with long-distance shooting with the Turkish bow, I append a letter written by one of my ancestors to another. They were both skilled and enthusiastic archers in their day. This letter, and the notes and translations which follow it, describe the extraordinary feats said to have been achieved by the Turks with their bows when shooting to attain a long range with a flight arrow:—

‘London, 1795.

‘Dear Brother,—I have just been to see the secretary of the Turkish Ambassador shooting with Waring⁵³ and other famous English bowmen. There was a great crowd, as you may suppose, to see them. The Turk,⁵⁴ regardless of the many persons standing round him, and to the amazement and terror of the Toxophilites, suddenly began firing his arrows up in all directions, but the astonishment of the company was increased by finding the arrows were not made to fly, but fell

harmlessly within a few yards. These arrows the Turk called his “exercising arrows.” This was an idea that was quite new to the bowmen present and they began to have more respect for the Turk and his bow. The Turk’s bow is made of antelopes’ horns and is short, and purposely made short for the convenience of being used in all directions on horseback.

‘The Toxophilites wished to see the powers of the Turkish bow, and the Turk was asked to shoot one of his flight arrows. He shot four or five, and the best flight was very carefully measured at the time. It was 482 yards. The Toxophilites were astonished, I can tell you.

‘Waring said the furthest distance attained with an English flight arrow, of which he had ever heard, was 335 yards, and that Lord Aylesford had once shot one, with a slight wind in his favour, 330 yards. Waring told me that he himself, in all his life, had never been able to send a flight arrow above 283 yards.

‘The Turk was not satisfied with his performance, but declared that he and his bow were stiff and out of condition, and that with some practice he could shoot very much further than he had just done.

‘He said, however, that he never was a first-class bowman, even when in his best practice, but that the present Grand Seigneur was very fond of the exercise and a very strong man, there being only two men in the whole Turkish army who could shoot an arrow as far as he could.

‘The Turk said he had seen the Grand Seigneur send a flight arrow 800 yards.

‘I asked Waring to what he attributed the Turk’s great superiority over our English bowmen; whether to his bow or not. Waring replied he did not consider it was so much the result of the Turk’s bow, but rather of his strength and skill, combined with the short light arrows he used, and his method of shooting them along the grooved horn attached to his hand.

‘Neither Waring nor any of the Toxophilites present (and many tried) could bend the bow as the Turk did when he used it.

‘So much for the triumph of the Infidels and the humiliation of Christendom.

‘Yours aff.,

‘W. FRANKLAND.

‘To Sir Thos. Frankland, Bt., M.P.
‘Thirkleby Park.’

⁵³ T. Waring, author of a *Treatise on Archery*, 1st ed. 1814, last ed. 1832. Waring was an accomplished archer and a well-known manufacturer of bows and arrows.

⁵⁴ Mahmoud Effendi.

I found the following records in a manuscript notebook of 1798, describing feats and incidents of archery collected by the recipient of the letter I have given.

‘Records of Turkish archery procured in 1797 from Constantinople by Sir Robert Ainslie, at the request of Sir Joseph Banks, and translated by Sir Robert Ainslie’s interpreter.

‘The Turks still have detachments of archers in their armies so as not to deviate from ancient custom, for in Turkey archery is now merely regarded as an amusing exercise that is to this day practised by all ranks of the people.

‘The Ottoman emperors, with their court, often enjoy the diversion of archery in public, and there is an extensive piece of ground allotted to that purpose.

‘This place is upon an eminence in the suburbs of the city of Constantinople and commands an extensive view of the town and harbour. It is called Ok Meydan, or the Place of the Arrow. The ground mentioned is covered with marble pillars erected in honour of those archers who have succeeded in shooting arrows to any remarkable distance. Each pillar is inscribed with the name of the person whose dexterity it records, together with some complimentary verses to him, and the exact range which he attained with his flight arrow.

‘The Ottoman emperors, from ancient times, have always been supposed to live by their manual labour and in consequence of this supposition they have each learnt some art or profession, most of them having preferred the art of making bows and arrows.

‘The present emperor was bound apprentice to the trade of archery, and at the time he was received as a master in this trade he gave very splendid public entertainments at the Ok Meydan, where the State tents were pitched for him and his court.’

Translations of the inscriptions on some of the marble columns at the Ok Meydan (Place of the Arrow) which were erected in honour of those who excelled in archery.

- | | | |
|--|-----|--------|
| 1. Ak Siraly Mustapha Aga shot two arrows both of which travelled to a distance of | 625 | yards. |
| 2. Omer Aga shot an arrow to a distance of | 628 | „ |
| 3. Seid Muhammed Effendy, son in-law of Sherbetzy Zade | 630 | „ |
| 4. Sultan Murad | 685 | „ |
| 5. Hagy Muhammed Aga shot an arrow | 729 | „ |
| 6. Muhammed Ashur Effendy shot an arrow which fixed in the ground at | 759 | „ |
| 7. Ahmed Aga, a gentleman of the Seraglio under Sultan Suleiman the Legislator, shot an arrow | 760 | „ |
| 8. Pashaw Oglee Medmed shot an arrow | 762 | „ |
| 9. The present Grand Admiral Hussein Pashaw shot an arrow which drove into the ground at | 764 | „ |
| 10. Pilad Aga, Treasurer to Hallib Pashaw | 805 | „ |
| 11. Hallib Aga | 810 | „ |
| 12. The reigning Emperor, Sultan Selim, shot an arrow which drove into the ground at a distance of | 838 | „ |
| The Sultan shot a second arrow to near the same distance. | | |

In the translation of the above from the Turkish language the feet and inches were also given for each shot, but these I have omitted as unnecessary.

In the manuscript, the interpreter remarks that the measurements of the distances on the marble columns at Ok Meydan are in pikes, the pike being a Turkish measure of a little over two feet, easily convertible into English yards, feet and inches.

It will be observed that the longest flight recorded on the columns selected for quotation is 838 yards, and the shortest, 625 yards. Though these distances are almost too extraordinary to be true, they corroborate the statement made in 1795 by the secretary of the Turkish ambassador, [p. 123](#). If they are correct, they can only be accounted for by the use of a light short arrow, a very powerful bow, great strength and skill, and above all else by the horn appendage which the Turkish archer attached to his left hand, and without which he could not shoot so short an arrow from his bow.

Even if we accept the shortest range recorded on the columns as correct—*i.e.* 625 yards—it is an extraordinary distance for any arrow to be propelled, and is 285 yards beyond what has ever been achieved, as far as we know, by an English bowman with a longbow, [p. 120](#).

It is, however, beyond question that the secretary to the Turkish Ambassador did shoot an arrow 482 yards (the arrow and bow being even now preserved in the Toxophilite Society's rooms), though he declared at the time of the occurrence that he was not proficient in the art of sending a flight arrow to what he considered a great distance. We may from this safely assume that a range of 143 yards further than the Turkish secretary attained with his bow, or a total flight of 625 yards, was quite possible in the case of a more powerful and skilled Turkish archer than he was.



TURKISH CAVALRY SOLDIERS WITH THEIR BOWS.

From an illuminated Turkish MS. in the Sloane Collection, B.M., dated 1621, No. 5258. These reproductions plainly show how small was the size of the bow formerly used in warfare by Turkish soldiers.

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